# SCIE3314: Agricultural Production Systems Simulator (APSIM)

## Introductory Tutorial

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**Modular Design for Crop Modelling and Decision Support**

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# Acknowledgments

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- APSOIL soil database (Australian soils)
- Weather files from BOM/SILO networks

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# Preface

This tutorial introduces APSIM (Agricultural Production Systems Simulator), a process-based crop modelling platform widely used in research, extension, and operational decision support.

**Learning Outcomes:** After completing this tutorial, students will be able to:

1. Understand the structure and daily sequence of APSIM simulations
2. Configure and run crop simulations for decision analysis
3. Interpret model outputs and assess crop responses to management
4. Apply APSIM to explore agronomic questions (N fertilisation, phenology, variability)
5. Troubleshoot common errors and validate simulation logic

## How to Use This Tutorial

**For Instructors:** Each module (0–7) is designed as a 30–90 minute class session or self-study unit. Four integrated exercises (A–D) scaffold hands-on simulation work, progressing from simple water balance to multi-year variability analysis. All exercises use near-solved .apsimx files (components pre-configured; students modify parameters).

**For Students:** Start with Module 0 (Why APSIM?), then progress sequentially through Modules 1–7. Complete Exercises A–D in parallel or after each module block. Each module includes key concepts, worked examples, and links to reference entities in the LLM Wiki (*see appendix F.7*).

**Prerequisites:** Basic agronomy (crop growth stages, soil water, nitrogen cycling). No programming required; JavaScript Manager rules are provided as templates.

# Chapter 1

# Module 0: Why APSIM?

## 1.1 Learning Objectives

By the end of this module, you will:

- Understand what crop modelling is and why it matters
- Identify key applications of APSIM in research, extension, and industry
- See APSIM in context alongside other simulation tools
- Recognize how APSIM connects to agronomic decision-making

## 1.2 What is Crop Modelling?

**Crop Modelling** is the use of mathematical equations and computational models to simulate crop growth, development, and yield in response to weather, soil, and management decisions.

### 1.2.1 Why Not Just Experiment?

Field experiments are essential but have limitations:

- **Time:** Experiments take seasons or years; models run in seconds
- **Cost:** Field trials are expensive; simulations are cheap
- **Scope:** Can't test every scenario; models explore hypotheticals
- **Climate:** Can't control weather; models test under different climates
- **Generalization:** Results from one site may not apply elsewhere; models extrapolate

Example: To test 5 sowing dates × 6 N rates × 3 varieties across 10 locations and 20 years of weather = 18,000 simulated scenarios. Field trials would take decades and millions of

dollars.

### 1.2.2 What Can Models Do?

Crop models allow farmers, agronomists, and researchers to:

- **Predict yield** under different management and climate scenarios
- **Optimize decisions** (N rate, sowing date, variety choice)
- **Assess risk** (variability across seasons; climate change impacts)
- **Understand mechanisms** (why does N affect grain protein?)
- **Train decisions support systems** (operational use in farms/regions)

## 1.3 APSIM: Overview

APSIM (Agricultural Production Systems Simulator) is a process-based crop modelling platform developed by:

- CSIRO (Australia)
- QAAFI (University of Queensland)
- CIMMYT (Mexico)
- ICRISAT (India)
- Michigan State University (USA)

### 1.3.1 Key Features

**Process-based** Models mechanistic interactions (photosynthesis, phenology, nutrient cycling), not just empirical curves

**Multi-component** Soil water, nitrogen, phosphorus, organic matter, crops, weather, management all coupled

**Daily timestep** High resolution for capturing short-term processes (daily rain, daily stress)

**Modular** Can mix-and-match components; run in research mode or farmer decision-support mode

**Open-source** Free; community-maintained; thousands of users worldwide

**Validated** Decades of testing against field data across 100+ crop types and locations

## 1.4 APSIM in Action: Real-World Applications

### 1.4.1 Research Example: Climate Change Impact on Wheat Yield

A researcher wants to know: *How will wheat yields change if rainfall decreases by 20% and temperatures increase by 2°C?*

1. Set up APSIM with current weather and soils for a wheat-growing region
2. Run baseline simulation (100 years of current climate)
3. Run scenario simulation (same 100 years but weather modified)
4. Compare yield distributions: baseline vs. scenario
5. Identify adaptation strategies (earlier sowing? different variety? more irrigation?)

### 1.4.2 Extension Example: Nitrogen Rate Optimization

An extension officer works with 50 wheat farmers in a district. Question: *What N rate maximizes profit for different soil types and seasons?*

1. Map farmer soils; input into APSIM
2. Run 5 N rates (0, 50, 100, 150, 200 kg/ha) $\times$ 20 years of regional weather
3. Calculate yield response and NUE (Nitrogen Use Efficiency)
4. Assess profit = Yield $\times$ Grain Price $-$ N Cost
5. Recommend N rate based on soil, season outlook, grain price

### 1.4.3 Operational Example: Sowing Decision Support

A farmer has a sowing window (May 1–July 10) and wants to know: *When should I sow to maximize yield?*

1. Set up APSIM with farm soil, local weather data (or forecast)
2. Sow on different dates; track emergence, phenology, water availability, final yield
3. Provide decision rule: "Sow when ESW $>$ 100 mm AND recent rain $>$ 25 mm"
4. Embed this rule in farm decision-support app; farmer checks conditions daily

## 1.5 APSIM Compared to Other Tools

**Table 1.1:** Comparison of Crop Modelling Platforms

| Feature | APSIM | DSSAT | RZWQM | AquaCrop |
|---|---|---|---|---|
| Process-based | Yes | Yes | Yes | Simplified |
| Multi-crop | 100+ | 40+ | 10+ | 30+ |
| N cycling | Yes | Yes | Yes | Limited |
| Climate datasets | Global | Global | Limited | Global |
| User interface | GUI + scripting | DOS/Web | Command-line | GUI |
| Cost | Free | Free | Free | Free |
| Learning curve | Steep | Moderate | Steep | Gentle |
| Active community | Large | Moderate | Small | Moderate |

**Note:** APSIM has a steeper learning curve but offers more flexibility and detail. AquaCrop is simpler but less mechanistic. DSSAT is similar to APSIM but different design philosophy. Choose based on your research question and available data.

## 1.6 Key Concepts in APSIM

### 1.6.1 Daily Simulation Loop

APSIM runs a daily timestep cycle:

1. **Clock advances** by one day
2. **Weather data** is read (temperature, rainfall, radiation)
3. **Manager evaluates rules** (is it time to sow? fertilise? harvest?)
4. **Soil processes** calculate water, N availability
5. **Crop model** calculates growth, N uptake, phenology
6. **Report** outputs selected variables
7. Repeat

### 1.6.2 Three-Layer Problem Representation

APSIM separates concerns:

**State layer** What is the current soil water, N pool, crop biomass, stage?

**Process layer** How do these states change? (PET, infiltration, phenology equations)

**Driver layer** What drives change? (Weather, management decisions)

This allows researchers to:

- Test process implementations in isolation
- Swap components (use different crop model, different soil model)
- Understand cause-effect pathways

## 1.7 Career Relevance

Crop modelling skills are in demand:

- **Research**: Universities, CSIRO, international agricultural research centers
- **Extension**: Government agencies, agricultural services, extension providers
- **Private sector**: Agribusiness, fertiliser companies, crop input providers
- **Policy**: Climate adaptation; food security; natural resource management
- **Operations**: Farm management software companies; decision support platforms

**Key Takeaway:** APSIM is a professional tool used globally in research, extension, and operations. Learning APSIM opens careers in agricultural science and decision support.

## 1.8 Roadmap for This Tutorial

This tutorial is structured in 8 modules:

**Module 0** Why APSIM? (this module) — motivation and overview

**Module 1** Structure — APSIM's component architecture

**Module 2** First Simulation — run a simple fallow water balance

**Module 3** Output Analysis — interpret simulation results

**Module 4** Nitrogen Fertilisation — explore N response curves

**Module 5** Phenology & Thermal Time — understand crop development

**Module 6** Long-Term Variability — assess climate risk

**Module 7** Climate Scenarios — explore adaptation strategies

Plus four hands-on exercises:

- **Exercise A**: Fallow water balance (7 years)
- **Exercise B**: N response factorial (4 rates)
- **Exercise C**: Sowing date factorial (3 scenarios)
- **Exercise D**: 20-year variability analysis

## 1.9 How APSIM Works: The Information Flow

**Process-Based Modelling:** A process-based model simulates the underlying biological and physical mechanisms (e.g., soil water flow, photosynthesis, nitrogen mineralisation). This contrasts with empirical models that fit statistical equations to observed data. Because APSIM represents mechanisms, it can be applied to conditions not included in its original calibration — making it suitable for evaluating novel management strategies or future climate scenarios.

Figure 1.1 summarises how information flows through an APSIM simulation each day.

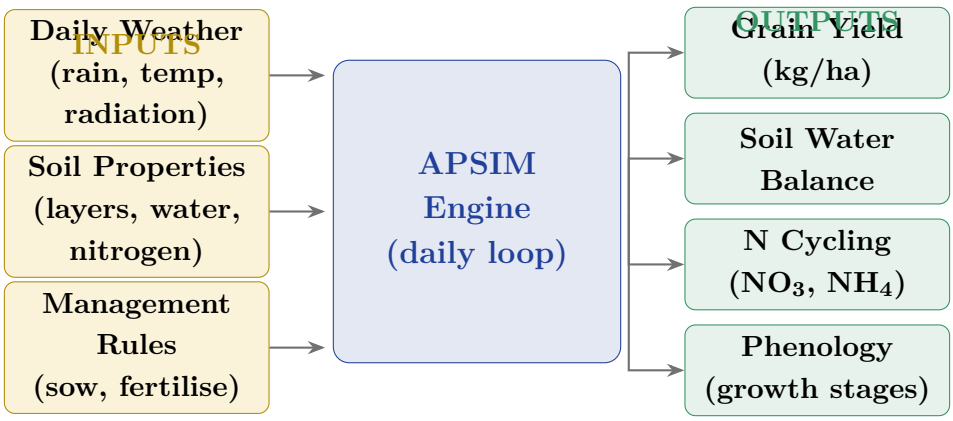


**Figure 1.1:** Conceptual overview of an APSIM simulation. Daily inputs drive a process-based calculation engine, generating outputs on crop yield, soil water, nitrogen cycling, and crop phenology.

## 1.10 Summary

- Crop models simulate crop-soil-weather-management interactions
- APSIM is a powerful, process-based platform with global adoption
- Applications span research, extension, and operational decision support
- This tutorial teaches APSIM through progressive modules + hands-on exercises
- Career pathways exist in research, extension, agribusiness, and policy

## 1.11 What's Next?

In Module 1, we dive into APSIM's structure: components, daily sequence, .apsimx file format, and how to interpret a simulation setup.

# Chapter 2

# Module 1: APSIM Structure

## 2.1 Learning Objectives

By the end of this module, you will:

- Understand APSIM's component hierarchy (Simulations → Simulation → Zone → Components)
- Recognize the daily execution sequence (Clock → Weather → Manager → Soil → Crop → Report)
- Interpret a `.apsimx` JSON file structure
- Understand the role of each component (Clock, Weather, Soil, Wheat, Manager, Report, DataStore)
- Navigate APSIM's GUI to view and modify simulation structure

## 2.2 APSIM's Component Architecture

### 2.2.1 Hierarchy

APSIM organizes all simulation information in a nested hierarchy:

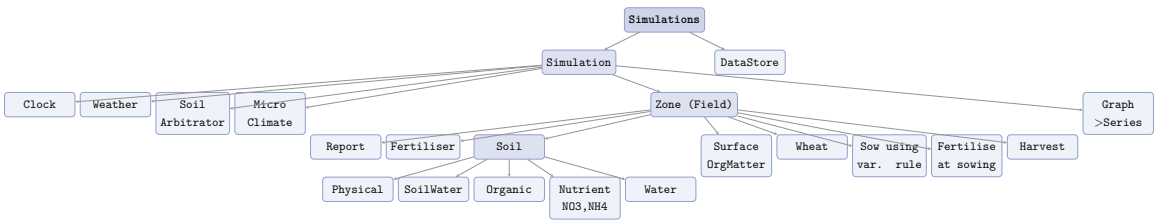


**Figure 2.1:** Component hierarchy of a typical APSIM wheat simulation. Each indented node is a child of the one above it — mirroring what you see in the APSIM left panel.

**Key Principle:** Components are pluggable. You can swap components (different crop model, different soil model) without changing the overall structure.

### 2.2.2 Global Components

**`Clock`** Advances simulation by one day. Sets start date, end date, timestep (always 1 day in APSIM). Synchronizes all other components to daily updates.

**`Weather`** Reads external weather file (`.met` format). Provides daily maximum temperature, minimum temperature, rainfall, radiation, humidity, wind speed to all other components.

**`SoilArbitrator`** Coordinates soil water and nutrient availability. When multiple processes access soil (e.g., plant root uptake, soil evaporation), arbitrator ensures consistency.

**`MicroClimate`** Calculates energy balance at soil surface. Partitions radiation (direct vs. diffuse), interception, albedo, reference height effects.

### 2.2.3 Zone Components

A Zone represents a spatial patch (typically 1 hectare for field simulations). Multiple zones allow side-by-side comparisons (e.g., different management, different soil).

**`Soil`** Multi-layer soil profile. Contains physical properties (bulk density, water characteristics), organic matter, chemical properties (pH, nutrients), and initial conditions. Sub-components: Physical, SoilWater, Organic, Chemical, Nutrient (N cycle), solutes ($NO_3$, $NH_4$, Urea), Temperature.

**`Wheat (or other Crop)`** Process-based crop model. Simulates phenology (thermal time, Zadok stages), biomass accumulation (photosynthesis, respiration), grain development, nutrient uptake, and stress responses.

**`Manager`** User-defined management rules. Written in JavaScript or C#; evaluated daily. Typical rules: sow when conditions met, apply fertiliser on date, harvest when physiologically mature.

**`Fertiliser`** Component that applies nutrient inputs. Manager calls `Fertiliser.Apply()` to add N, P, K at specified times.

**`Report`** Output filter. Specifies which variables APSIM saves to the database. Reduces output file size by excluding irrelevant variables.

**`DataStore`** Database (SQL-like) that receives daily reports and exports to Excel/CSV at simulation end.

## 2.3 Daily Execution Sequence

Each day, APSIM follows a strict sequence:

1. **Clock.Today** advances by 1 day
2. **Weather** reads data for today
3. **Manager** evaluates and executes all rules
4. **Soil** calculates water balance (infiltration, drainage, evaporation)
5. **Wheat** calculates phenology, photosynthesis, N uptake, growth
6. **Report** outputs selected variables to database
7. **DataStore** receives report data
8. Repeat (next day)

**Important:** Order matters! Manager runs before Soil/Crop, so today's sowing decision uses yesterday's soil state. Soil runs before Crop, so water and N availability affect today's crop growth. This sequence is fundamental to APSIM.

## 2.4 The .apsimx File Format

APSIM simulations are stored as JSON files (`.apsimx`). Example structure:

```
{
  "$type": "Models.Core.Simulations",
  "Name": "Simulations",
  "Children": [
    {
      "$type": "Models.Core.Simulation",
      "Name": "Simulation1",
      "Children": [
        {
          "$type": "Models.Clock",
          "Start": "1900-01-01",
          "End": "2000-12-31",
          "Name": "Clock"
        },
        {
```

```
          "$type": "Models.Climate.Weather",
          "FileName": "%root%/Examples/WeatherFiles/AU_Northam.
             met",
          "Name": "Weather"
        },
        { ... more components ... }
      ]
    },
    {
      "$type": "Models.Storage.DataStore",
      "Name": "DataStore"
    }
  ]
}
```

**Listing 2.1:** Simplified .apsimx structure

**Key Observation:** The JSON structure mirrors the component hierarchy. APSIM reads this file at startup, instantiates components, and establishes linkages. You can edit `.apsimx` files directly in a text editor, but APSIM's GUI is more user-friendly.

## 2.5 Component Reference Table

**Table 2.1:** APSIM Components: Role and Typical Configuration

| Component | Type | Configuration |
|---|---|---|
| Clock | Input | Dates (start, end), timestep |
| Weather | Input | File path (`.met`), variables |
| Soil | State | Layers, properties, initial water/N |
| Wheat | Process | Phenology, LAI, grain fill |
| Manager | Control | Rules (JavaScript or C#) |
| Report | Output | Variable list |
| DataStore | Output | Database (auto-populated) |

## 2.6 Interpreting a Real Simulation: wheat_example.apsimx

We'll use `wheat_example.apsimx` (provided in `Apsim_examples/`) as a reference model.

### 2.6.1 Key Configuration

**Location** Northam, Western Australia (Red/Brown earth soil)

**Climate** 101 years (1900–2000) from SILO/BOM weather file

**Crop** Wheat (Hartog cultivar), 120 plants/m$^2$

**Soil** 7-layer profile, plant-available water = 361 mm

**Management** 3 rules: sow (ESW > 100 mm + rain), fertilise (160 kg/ha N at sowing), harvest (at maturity)

### 2.6.2 Output Variables

Report tracks 13 daily variables:

- Phenology: Zadok stage, stage name
- Growth: LAI, total biomass, grain mass
- Grain quality: Grain number, size, protein, N content
- Yield: Grain total weight

## 2.7 Common Misconceptions

**Misconception 1:** "I can change timestep from daily to hourly."
*Reality:* APSIM is designed for daily timestep only. Many processes (phenology, root growth) are not parameterized for sub-daily resolution. If you need hourly resolution, consider alternative models (e.g., DSSAT in hourly mode, though rare).

**Misconception 2:** "Manager runs continuously, not daily."
*Reality:* Manager evaluates once per day (during the daily sequence). If you want multiple decisions per day (e.g., separate sowing and fertiliser decisions), both should be Manager rules that fire on the same day (order determined by rule sequence).

**Misconception 3:** "I can edit `.apsimx` files without understanding JSON."
*Reality:* JSON is strict. Mismatched braces, quotes, or commas will cause APSIM to fail. Use APSIM's GUI to create files; edit manually only if you understand JSON syntax or use a JSON validator.

## 2.8 Navigating APSIM's GUI

### 2.8.1 Opening a Simulation

1. Launch APSIM
2. File → Open → select `.apsimx` file
3. You should see a tree structure on the left (Components), properties panel on the right

### 2.8.2 Viewing Components

- Click on a component in the tree to view its properties
- Properties show all configurable parameters (dates, N rate, etc.)
- Some properties have dropdown menus (e.g., cultivar names, soil names)

### 2.8.3 Running a Simulation

1. Select the Simulation in the tree
2. Click the "Run" button (or Ctrl+R)
3. APSIM displays progress in the output window
4. When complete, click "Generated Outputs" or "DataStore" to view results

### 2.8.4 Exporting Results

1. Right-click the DataStore in the tree
2. Select "Export" → "To Excel" or "To CSV"
3. Choose filename and location
4. Results are now ready for analysis in Excel or Python

## 2.9 Summary

- APSIM structure: Hierarchy of components (Clock, Weather, Soil, Crop, Manager, Report, DataStore)
- Daily sequence: Clock → Weather → Manager → Soil → Crop → Report
- .apsimx file is JSON; mirrors component hierarchy

- Each component has specific role and configuration
- APSIM GUI provides user-friendly interface for viewing and modifying simulations

## 2.10 Further Resources

**Note:** Detailed component reference pages are planned for the APSIM tutorial online knowledge base. For now, consult the official APSIM Next Gen documentation at:

https://apsimnextgeneration.netlify.app/

Key topics to explore: Clock (simulation timing), Weather (climate input), Soil (profile configuration), Wheat (crop model), Manager (management rules), Report (output variables), and DataStore (results database).

## 2.11 What's Next?

In Module 2, you'll set up and run your first APSIM simulation: a simple fallow water balance.

# Chapter 3

# Module 2: First Simulation

## 3.1 Learning Objectives

By the end of this module, you will:

- Set up a simple APSIM simulation from scratch
- Understand fallow water balance: rainfall, runoff, drainage, evaporation
- Interpret daily ESW (extractable soil water) as an indicator of sowing readiness
- Run simulation and export results to Excel
- Troubleshoot common startup errors (file not found, missing dates, etc.)

## 3.2 Conceptual Framework: Fallow Water Balance

### 3.2.1 What is a Fallow?

A fallow is a season where the field is left without a crop. Soil water accumulates (useful before sowing) or is lost to drainage/evaporation.

**Daily Water Balance Equation:**

$$\text{ESW}_t = \text{ESW}_{t-1} + \text{Rainfall} - \text{Runoff} - \text{Drainage} - \text{Evaporation}$$

where ESW = extractable soil water (mm), available for plant growth.

### 3.2.2 When Are Soils Ready for Sowing?

Farmers typically wait for:

- ESW > 100 mm (soil has adequate water to support germination and early growth)
- Rainfall > 25 mm in last 7 days (recent rain indicates favorable conditions)

- Date within sowing window (e.g., May 1 – July 10 for winter wheat in Queensland)

Exercise A explores these thresholds using a 7-year fallow simulation.

## 3.3 Getting Started: Open Exercise A

### 3.3.1 File Location

`Exercise_A_Fallow_Water_Balance.apsimx` (provided in course materials)

### 3.3.2 Simulation Setup

**Clock:** 7 years of a representative site (e.g., 1950–1957)

**Weather:** Northam, Western Australia (real BOM/SILO data)

**Soil:** Red/Brown earth (shallow to medium depth; realistic for WA Wheatbelt wheat region)

**Management:** None (fallow = no crop, no fertiliser)

**Report Variables:**

Clock .Today (date)

Soil .ESW (extractable soil water)

Weather .Rainfall (daily rain)

Soil .SoilWater.Evaporation (simulated evaporation)

Soil .SoilWater.Drainage (deep percolation)

## 3.4 Running the Simulation

### 3.4.1 Step 1: Open APSIM

1. Launch APSIM Next Gen
2. File → Open → select `Exercise_A_Fallow_Water_Balance.apsimx`

### 3.4.2 Step 2: View Structure

- Expand the tree on the left
- Verify components: Clock, Weather, Soil, Report, DataStore

- Click each component to check configuration (dates, file paths, variables)

### 3.4.3 Step 3: Run Simulation

1. Click the Simulation in the tree
2. Press Run button (or Ctrl+R)
3. Watch the output window for progress
4. Simulation should complete without errors

### 3.4.4 Step 4: View Results

1. Click DataStore in the tree
2. Right-click → Export → To Excel
3. Name file: `Exercise_A_Results.xlsx`
4. Open in Excel; examine data

## 3.5 Analyzing the Data

### 3.5.1 Key Columns

**Table 3.1:** Output Variables in Exercise A

| Column | Units | Interpretation |
|---|---|---|
| Clock.Today | Date | Calendar date |
| Soil.ESW | mm | Plant-available water; sowing threshold is ESW > 100 mm |
| Weather.Rainfall | mm | Daily precipitation |
| Soil.SoilWater.Evaporation | mm | Soil water lost to atmosphere (PET if no crop) |
| Soil.SoilWater.Drainage | mm | Water percolating below root zone |

### 3.5.2 Analysis Prompts

1. **Seasonal Pattern:** Plot ESW vs. date across all 7 years. When does ESW peak? When is it lowest?

2. **Sowing Window:** Identify dates when ESW > 100 mm AND rainfall in last 7 days > 25 mm. How many sowing opportunities per year?

3. **Dry Year vs. Wet Year:** Compare ESW profiles for a dry year vs. wet year. What drives the difference?

4. **Evaporation vs. Drainage:** Which process removes more water? Does it vary seasonally?

## 3.6 Common Errors and Troubleshooting

**Error 1: "File not found" for weather file**
*Solution:* Weather file path uses `%root%` variable (APSIM's installation directory). Verify weather file exists at: `%APSIM_Installation%/Examples/WeatherFiles/AU_Northam.met`. If not found, check APSIM installation or download file separately.

**Error 2: "Soil profile empty" or "missing layers"**
*Solution:* Ensure Soil component has Physical, SoilWater, and Nutrient sub-components. These are auto-created if you load a predefined soil from APSOIL database. If manually editing `.apsimx`, verify JSON structure is correct.

**Error 3: "Report variables not found"**
*Solution:* Verify variable names match APSIM's internal paths. Example: `[Soil].ESW` not `Soil.ESW`; `[Weather].Rainfall` not `Weather.Rain`. Consult APSIM GUI (right-click Report) to auto-generate valid variable names.

## 3.7 Extending the Analysis

### 3.7.1 Comparison: Different Soils

Create two variants of Exercise A:

1. Same as above (black clay)

2. Shallow sandy soil (low water-holding capacity)

Run both; compare ESW profiles. Which soil reaches sowing threshold more often?

### 3.7.2 Comparison: Different Weather

Use a different weather file (e.g., wetter/drier region). How does ESW dynamics change?

### 3.7.3 Multi-Year Statistics

Calculate for all 7 years:

- Average ESW by month
- Frequency of sowing-ready days
- Coefficient of variation (year-to-year variability)

## 3.8 Conceptual Questions

1. Why is ESW in the fallow higher in spring than summer, even if rainfall is similar?
2. If you sow wheat on day X when ESW = 110 mm, does ESW stay above 100 mm? Why or why not?
3. How would deep drainage change if the soil were twice as thick (double the LL and DUL)?
4. Could a rainy day during grain fill harm wheat yield? How would APSIM capture this?

## 3.9 Summary

- Fallow water balance is foundational: daily rain/runoff/drainage/evaporation
- Sowing readiness depends on ESW and recent rain (decision threshold)
- Exercise A provides 7-year dataset for analyzing water availability
- Results can be exported to Excel for custom analysis and visualization
- Comparing soil types and weather regions reveals biophysical constraints

## 3.10 Soil Water Reference

### 3.10.1 Soil Water Thresholds

APSIM defines five key water-content thresholds for each soil layer (Table 3.2).

**Table 3.2:** Soil water-content thresholds used in APSIM.

| Full name | Symbol | Meaning |
|---|---|---|
| Air Dry | `AirDry` | Minimum water content (oven-dry equivalent). |
| Wilting Point | `LL15` | Water content at 1500 kPa; plants cannot extract water below this threshold. |
| Field Capacity | `DUL` | Drained Upper Limit; water content after free drainage has ceased (≈10 kPa). |
| Saturation | `SAT` | Maximum water content; all pores filled. |
| Crop Lower Limit | `Wheat LL` | Minimum water extractable by wheat roots; may differ from LL15 in subsoil layers. |

**Extractable Soil Water (ESW):** For each layer:

$$\text{ESW (mm)} = (\text{SW} - \text{Wheat LL}) \times \text{layer thickness (mm)}$$

where SW is the current volumetric water content. Summed across all layers, `sum([Soil].SoilWater.ESW)` gives the total plant-available water in the profile (mm). This is a key indicator of crop water supply and the trigger variable for sowing decisions.

### 3.10.2 Soil Water Balance

The daily water balance equation describes how water enters, moves through, and leaves the soil profile (Figure 3.1):

$$\Delta\,\text{ESW} = P - R_{\text{off}} - D - E_s - T_c \tag{3.1}$$

where $P$ = precipitation, $R_{\text{off}}$ = runoff, $D$ = drainage below the root zone, $E_s$ = soil evaporation, and $T_c$ = crop transpiration.

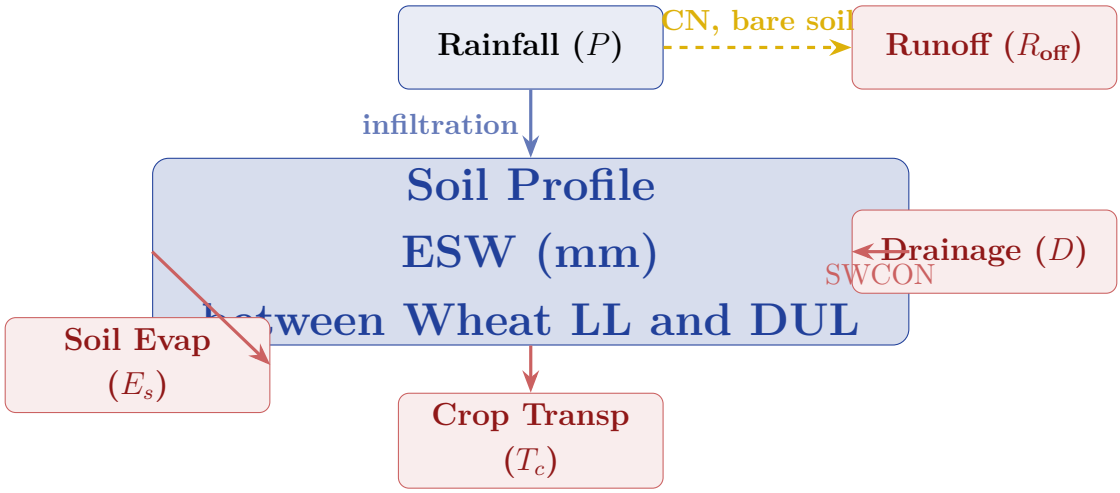


**Figure 3.1:** Soil water balance in APSIM. Rainfall either runs off (controlled by CN) or infiltrates into the soil profile. Stored ESW is depleted by drainage, soil evaporation, and crop transpiration.

### 3.10.3 Water Balance Report Variables

**Table 3.3:** Key APSIM report variables for soil water balance analysis.

| Variable | Units | Description |
|---|---|---|
| `sum([Soil].SoilWater.ESW)` | mm | Total extractable soil water (all layers). |
| `[Soil].SoilWater.Runoff` | mm/day | Surface runoff leaving the paddock. |
| `[Soil].SoilWater.Drainage` | mm/day | Water draining below the root zone. |
| `[Soil].SoilWater.Es` | mm/day | Soil surface evaporation. |
| `[Wheat].WaterUptake` | mm/day | Crop water uptake (transpiration). |
| `[Weather].Rain` | mm/day | Daily rainfall. |

## 3.11 Further Resources

**Note:** Additional reference material on ESW (Extractable Soil Water), Water Balance, Soil profiles, and Manager logic is available in the APSIM Next Gen documentation. For sowing rules and advanced water balance concepts, refer to Appendices A and B of this tutorial.

## 3.12 What's Next?

In Module 3, you'll learn to interpret simulation outputs: calculating averages, plotting trends, and extracting agronomic insights from data tables.

# Chapter 4

# Module 3: Output Analysis

## 4.1 Learning Objectives

By the end of this module, you will:

- Locate and export simulation results from the APSIM `DataStore`
- Import APSIM output into Excel and Python
- Create summary statistics (mean, min, max, standard deviation)
- Generate time-series and bar plots from daily output
- Interpret crop growth curves (LAI, biomass, grain mass phases)
- Calculate key agronomic metrics: yield, NUE, and Harvest Index

## 4.2 APSIM Output Structure

Each row in an APSIM output table represents one simulated day. Columns correspond to the variables you selected in the `Report` component.

| Date | ESW (mm) | Rainfall (mm) | Drainage (mm) | Evaporation (mm) |
|---|---|---|---|---|
| 1950-01-01 | 250 | 0 | 0 | 3.5 |
| 1950-01-02 | 246 | 0 | 0 | 4.1 |
| 1950-01-03 | 257 | 15 | 0 | 2.1 |
| ... | ... | ... | ... | ... |

**Output = daily snapshot.** Every column is a state variable recorded at the end of that day's calculation cycle. If a variable is missing from your output, check the `Report` component's variable list and re-run the simulation.

### 4.2.1 Variable Naming Convention

APSIM uses dot notation to identify variables:

```
[ComponentName].[SubComponent].[Variable]

Example:  [Soil].SoilWater.ESW
Example:  [Wheat].GrainMass
Example:  [Weather].Rainfall
```

The square brackets identify the component name; the dot separates levels of the hierarchy. These names appear as column headers in exported data.

## 4.3 Exporting Data from APSIM

### 4.3.1 Step-by-Step: Export to Excel

1. Run your simulation (press **Run** or F5)
2. In the `Simulations` tree on the left, click your simulation node
3. In the right-hand panel, click the **Results** tab at the bottom of the screen
4. Right-click anywhere in the results table
5. Select **Export** → **Excel (.xlsx)**
6. Save the file to your working folder

*[IMAGE: Screenshot of APSIM with the Results tab selected and the right-click Export menu visible.*
*Annotate: (1) Simulation tree, (2) Results tab at bottom, (3) Right-click Export option]*

**Figure 4.1:** Exporting simulation output from the APSIM DataStore. Select the Results tab, right-click the table, and choose Export.

**CSV alternative:** If Excel is not available, select **CSV (.csv)** instead. CSV files open in any text editor and import cleanly into Python, R, or Jupyter without modification.

## 4.4 Excel Analysis

### 4.4.1 Opening the Exported File

1. Open the exported `.xlsx` file in Excel
2. Confirm the sheet name matches your simulation name
3. Check the first and last rows for date range (should match your Clock start/end dates)
4. Scan the column headers — they should match the variable names from your `Report`

**Common issue:** The Date column may import as plain text rather than a date. *Fix:* Select the Date column → Data → Text to Columns → select Date (YMD) → Finish.

### 4.4.2 Summary Statistics with Excel Formulas

Once your data is loaded, calculate basic statistics for any numeric column. For a column in cells B2:B2558 (seven years of daily data):

| Statistic | Excel Formula |
|---|---|
| Mean | `=AVERAGE(B2:B2558)` |
| Minimum | `=MIN(B2:B2558)` |
| Maximum | `=MAX(B2:B2558)` |
| Std Deviation | `=STDEV(B2:B2558)` |
| Days with Rainfall > 0 | `=COUNTIF(B2:B2558,">0")` |
| Days with ESW > 100 mm | `=COUNTIF(C2:C2558,">100")` |

**Tip:** Create a separate summary sheet. Reference raw data in that sheet (e.g., `=Sheet1!B2:B2558`) so your original data stays unchanged and the summary is easy to update when you switch simulations.

### 4.4.3 Monthly Averages Using a PivotTable

1. Click anywhere in your data table
2. Select **Insert** → **PivotTable** → **New Worksheet**

3. In the PivotTable field list:
   - Drag **Date** to the **Rows** area
   - Right-click any date cell in the table → **Group** → tick **Months** and **Years** → OK
   - Drag **ESW** (or your variable of interest) to the **Values** area
   - In the Values area, click the drop-down arrow → **Value Field Settings** → choose **Average**
4. You now have average ESW for each month-year combination

*[IMAGE: Excel screenshot showing a PivotTable with Date (grouped by Month/Year) in rows and "Average of ESW" in values.*
*Annotate: (1) Field list panel, (2) Date grouping dialog, (3) Resulting summary table]*

**Figure 4.2:** Excel PivotTable for monthly average ESW. Group the Date field by Month and Year to produce seasonal summaries without writing formulas.

### 4.4.4 Visualisation in Excel

**Time-Series Line Chart (ESW Over 7 Years)**

1. Select the Date column, then hold Ctrl and select the ESW column
2. Select **Insert** → **Charts** → **Line** → **Line**
3. Add a title: "Fallow Water Balance: ESW 1950–1956"
4. Label axes: X = "Date", Y = "ESW (mm)"
5. Add a reference line at ESW = 100 mm (sowing threshold):
   - In a spare column, enter 100 for every row
   - Right-click the chart → **Select Data** → **Add Series**
   - Select the column of 100s; set series name to "Sowing threshold"
   - Format this series as a dashed red line with no markers

**Annual Rainfall Bar Chart**

1. In a spare area, create a table with years (1950, 1951, . . . , 1956) in column A
2. In column B, calculate annual rainfall using:
   `=SUMPRODUCT((YEAR($A$2:$A$2558)=D2)*$B$2:$B$2558)`
   where D2 contains the year value
3. Select the year and rainfall columns → Insert → Column Chart

**Conditional formatting:** Highlight sowing-opportunity rows automatically. Select the ESW column → Home → Conditional Formatting → Highlight Cell Rules → Greater Than → enter 100 → choose green fill.

## 4.5 Python Analysis

**Interactive Notebook Available:** A complete Jupyter notebook with all Python analysis code is provided as `APSIM_Analysis_Notebook.ipynb`. Students can:

- Open it directly in Google Colab (no installation required)
- Upload APSIM CSV exports and run analysis interactively
- Ask AI assistants (Gemini, ChatGPT, Claude) for help with code or interpretation
- Modify code to suit their specific simulation scenarios

Access the notebook from the course LMS or GitHub repository.

### 4.5.1 Reading Data

```python
import pandas as pd
import matplotlib.pyplot as plt

# Read exported CSV
df = pd.read_csv('Exercise_A_Results.csv')

# Convert date column
df['Date'] = pd.to_datetime(df['Date'])

# Examine first few rows
print(df.head())
```

**Listing 4.1:** Read APSIM output in pandas

### 4.5.2 Summary Statistics

```
# Extract year and month
df['YearMonth'] = df['Date'].dt.to_period('M')

# Monthly average ESW
monthly_esw = df.groupby('YearMonth')['ESW'].mean()
print(monthly_esw)
```

**Listing 4.2:** Calculate monthly averages

### 4.5.3 Visualization

```
plt.figure(figsize=(12, 5))
plt.plot(df['Date'], df['ESW'], linewidth=1)
plt.axhline(y=100, color='r', linestyle='--', label='Sowing
    threshold')
plt.xlabel('Date')
plt.ylabel('ESW (mm)')
plt.title('Fallow Water Balance: 7-Year ESW Trend')
plt.legend()
plt.grid()
plt.show()
```

**Listing 4.3:** Plot ESW time series

## 4.6 Agronomic Metrics

### 4.6.1 Yield

APSIM reports `[Wheat].GrainMass` in kg/ha at harvest. Use this value directly. Typical dryland wheat yields in subtropical Queensland: 1500–4500 kg/ha depending on rainfall.

**Note:** Some APSIM versions report grain mass in g/m$^2$. To convert to kg/ha: multiply by 10. Always check the Report variable description if yield values seem unexpectedly high or low.

### 4.6.2 Nitrogen Use Efficiency (NUE)

$$\text{AEN} = \frac{\text{Yield}_{N\ \text{treatment}} - \text{Yield}_{0\ \text{N}}}{\text{N rate}}$$

Typical range: 10–20 kg grain per kg N applied

### 4.6.3 Harvest Index

$$\text{HI} = \frac{\text{Grain mass}}{\text{Total aboveground biomass}}$$

Typical wheat: 0.4–0.5 (40–50% of biomass is grain)

## 4.7 Interpreting Crop Growth Curves

When you plot daily crop variables against date, four distinct phases become visible:

| Phase | Zadok stages | What you see in output |
|---|---|---|
| Establishment | 0–19 | LAI rises from 0; Biomass near zero |
| Vegetative | 20–49 | LAI increases rapidly; Biomass growing |
| Reproductive | 50–69 | LAI at maximum; Biomass growth peaks |
| Grain fill | 70–89 | LAI declining; GrainMass increases sharply |
| Maturity | 90+ | LAI = 0; GrainMass plateaus (harvest triggered) |

*[IMAGE: Dual-axis plot of APSIM output: Biomass (kg/ha, left axis) and LAI ($m^2/m^2$, right axis) vs Date.*
*Annotate key phases with vertical dashed lines: Emergence, Anthesis, Grain fill start, Maturity]*

**Figure 4.3:** Typical wheat crop growth curve from APSIM. Biomass grows continuously until maturity; LAI peaks around anthesis then declines as leaves senesce.

**Use the Zadok stage column as a guide.** In your exported data, find rows where `[Wheat].Stage` ≈ 60 (anthesis) and `[Wheat].Stage` ≈ 70 (grain fill start). Check that LAI is near its peak at anthesis and that GrainMass is essentially zero until grain fill begins. If either is missing, re-check your Manager harvest rules and Report variables.

## 4.8 Common Analysis Mistakes

**Mistake 1:** Comparing simulations with different date ranges
*Solution:* Normalize to common period or use per-season metrics (e.g., cumulative rainfall in growing season, not absolute date ranges)

**Mistake 2: Not accounting for missing values before sowing**
APSIM reports `NaN` or blank for crop variables (e.g., `[Wheat].GrainMass`) during the fallow period before the crop is sown. Averaging a column that includes these blanks will underestimate the true mean.
*Fix:* In Excel, use `AVERAGEIF` to exclude blanks. In Python, use `df['col'].dropna().mean()`.

**Mistake 3: Interpreting daily output as seasonal totals**
The `Rainfall` column is the rainfall on that day (mm/day), not cumulative. Similarly, ESW is the soil water on that day, not total water added.
*Fix:* To get seasonal totals, use `SUM` (or `groupby().sum()` in Python) on the relevant date range.

## 4.9 Summary

**Module 3 key points:**

1. APSIM output = one row per simulated day; columns = `Report` variables
2. Export from `DataStore`: Results tab → right-click → Export → Excel or CSV
3. Excel: verify date format → PivotTable for monthly averages → line chart for time series
4. Python: `pd.read_csv()` → `groupby()` → `matplotlib`
5. Core metrics: Yield (kg/ha), AEN (kg grain / kg N), Harvest Index (0.4–0.5 for wheat)
6. Plot crop growth curves and use Zadok stage to verify all phases are present

## 4.10 What's Next?

Module 4 explores nitrogen fertilisation and N response curves.

# Chapter 5

# Module 4: Nitrogen Fertilisation

## 5.1 Learning Objectives

By the end of this module, you will:

- Understand N cycling in APSIM (soil N pools, uptake, plant demand)
- Generate N response curves (yield vs. N rate)
- Calculate NUE (Nitrogen Use Efficiency)
- Assess N stress effects on crop development and protein
- Optimize N rate based on cost-benefit analysis

## 5.2 Nitrogen Cycling in APSIM

### 5.2.1 Soil N Pools

- **$NO_3$ (nitrate):** Mobile; easily leached; available to plant
- **$NH_4$ (ammonium):** Less mobile; adsorbed to clay; slowly nitrified
- **Organic N:** Immobilized in soil organic matter; mineralized slowly
- **Urea:** Applied as fertiliser; hydrolyzed to $NH_4$ over days

### 5.2.2 Daily N Processes

1. **Mineralization:** Organic N $\rightarrow$ $NH_4$
2. **Nitrification:** $NH_4 \rightarrow NO_3$
3. **Plant uptake:** N availability $\times$ plant demand = actual uptake
4. **Leaching:** $NO_3$ lost with drainage water
5. **Immobilization:** N incorporated into new microbial biomass

## 5.3 N Response Curves

### 5.3.1 Typical Response

When N rate increases from 0 to a high rate, yield follows a diminishing-returns curve:

1. At **low N** (0–80 kg/ha): yield increases steeply — N is strongly limiting
2. At **moderate N** (80–160 kg/ha): yield still increases but more slowly
3. At **high N** (>160 kg/ha): yield approaches a plateau — other factors limit (water, light, genetics)

Typical wheat N response at Northam (Red/Brown earth, Mediterranean WA Wheatbelt):

| N Rate (kg/ha) | Yield (kg/ha) | Yield gain (kg/ha) | NUE (kg/kg) |
|---:|---:|---:|---:|
| 0 | 1800 | — | — |
| 80 | 2960 | 1160 | 14.5 |
| 160 | 3780 | 820 | 12.4 |
| 240 | 4060 | 280 | 9.4 |

**Note:** The values above are indicative. Your simulated values will differ depending on the season's rainfall and initial soil N. In Exercise B you will calculate these numbers from your own simulation output.

*[IMAGE: N response curve plot. X-axis: N rate (0, 80, 160, 240 kg/ha). Y-axis: Grain yield (kg/ha). Show diminishing returns shape. Mark optimal economic N rate with vertical dashed line.]*

**Figure 5.1:** Typical wheat N response curve showing diminishing returns at high N rates. The optimal N rate depends on grain price and fertiliser cost.

### 5.3.2 Exercise B: N Response Factorial

Four N rate variants:

- 0 kg/ha (control)

- 80 kg/ha
- 160 kg/ha (standard)
- 240 kg/ha (high)

Run all four; compare yields, grain protein, and NUE.

## 5.4 Nitrogen Use Efficiency

$$\text{NUE} = \frac{\text{Yield}_{N\ \text{treatment}} - \text{Yield}_{0\ \text{N}}}{\text{N rate}} \quad [\text{kg grain / kg N}]$$

**Interpretation:**

- NUE = 15 kg/kg: Every kg of N produces 15 kg of grain
- NUE is typically highest at low N rates (diminishing returns)
- NUE decreases as N rate increases (physiological limit)

## 5.5 N Stress and Crop Responses

When soil N supply cannot meet crop demand, APSIM applies an N stress factor that reduces growth:

| Stressed component | Effect | Visible in output |
|---|---|---|
| Leaf area (LAI) | Reduced leaf expansion | Lower LAI peak |
| Biomass accumulation | Slower carbon fixation | Lower biomass at anthesis |
| Grain number | Fewer florets set | Lower grain count at Stage 65 |
| Grain protein | Higher protein if grain N > carbohydrate | `[Wheat].GrainProtein` rises |
| Harvest Index | Grain fill incomplete | HI < 0.40 |

**N stress vs water stress:** Both reduce yield, but through different mechanisms.

- **N stress**: Reduces leaf area and biomass accumulation (fewer sources of carbon)
- **Water stress**: Reduces stomatal conductance and transpiration (fewer sinks fulfilled)
- In APSIM, the more limiting stress dominates on any given day

## 5.6 Cost-Benefit Analysis

### 5.6.1 Profit Formula

Net Profit (\$/ha) = (Yield (kg/ha)×Grain price (\$/kg))−(N rate (kg/ha)×N cost (\$/kg N))

Typical values for Queensland dryland wheat (indicative only):

- Grain price: \$0.28–0.35 per kg (\$280–350 per tonne)
- Urea (46% N): approximately \$1.20–1.60 per kg N (varies with market)
- Application cost: approximately \$15–25 per ha

### 5.6.2 Worked Example

Using the indicative N response values from Section 4.3:

| N Rate (kg/ha) | Yield (kg/ha) | Revenue @ \$0.30/kg | N Cost @ \$1.40/kg N | Net Profit (\$/ha) |
|---|---|---|---|---|
| 0 | 1800 | 540 | 0 | 540 |
| 80 | 2960 | 888 | 112 | 776 |
| 160 | 3780 | 1134 | 224 | 910 |
| 240 | 4060 | 1218 | 336 | 882 |

In this example, 160 kg/ha gives the highest net profit. At 240 kg/ha, the additional yield does not justify the extra fertiliser cost.

**Year-to-year variability matters.** In a dry year, yield potential is lower, so the optimal N rate is lower. In Exercise B you will compare profit across a single season. Exercise D extends this to 20 years to reveal how profit varies with climate.

## 5.7 Regional Variability and AEN Interpretation

AEN varies significantly across regions because it depends on soil N mineralisation, rainfall, and variety:

| Region type | Typical AEN (kg grain/kg N) | Key driver |
|---|---|---|
| High-rainfall, leaching soil | 8–12 | N lost to drainage; mode |
| Subtropical clay soil (Northam, WA) | 8–12 | Variable N retention; mo |
| Drought-prone, low rainfall | 6–10 | Water limits yield regard |
| High-mineralising soil | 8–12 | Soil supplies some N; cro |

**Why WA Red/Brown earth soils respond to N:** The Red/Brown earth soil at Northam has moderate cation-exchange capacity, which adsorbs $NH_4^+$ and reduces leaching. However, it drains more freely than Black Vertosols, so N management requires attention to timing and rainfall. NUE on these soils is typically moderate (10–15 kg/kg), reflecting both the soil's capacity to retain applied N and the constraints of the Mediterranean climate.

## 5.8 Summary

**Module 4 key points:**

1. Soil N exists in four pools: $NO_3^-$, $NH_4^+$, organic N, and urea; only $NO_3^-$ and $NH_4^+$ are immediately plant-available
2. Daily N processes: mineralisation → nitrification → plant uptake → leaching
3. N response curves follow diminishing returns: large yield gains at low N, small gains at high N
4. NUE = ($\text{Yield}_N$ − $\text{Yield}_0$) / N rate; typical range 10–16 kg grain/kg N for dryland wheat
5. N stress reduces LAI, biomass, grain number, and Harvest Index; raises grain protein
6. Optimal N rate depends on grain price, fertiliser cost, and seasonal rainfall — use cost-benefit analysis

## 5.9 What's Next?

Module 5 covers phenology and thermal time (how development depends on temperature).

# Chapter 6

# Module 5: Phenology & Thermal Time

## 6.1 Learning Objectives

By the end of this module, you will:

- Understand thermal time (Growing Degree Days) as driver of crop development
- Recognize Zadok stages (0–90+) as standardized phenological scale
- Explain why temperature, not calendar, matters for development
- Use Zadok stages to identify critical periods (booting, grain fill)
- Apply sowing date effects to grain quality and yield

## 6.2 Thermal Time: Concept and Calculation

### 6.2.1 Growing Degree Days (GDD)

$$\mathrm{GDD}_{\mathrm{daily}} = \max(0, T_{\mathrm{avg}} - T_{\mathrm{base}})$$

where:

- $T_{\mathrm{avg}} = (\mathrm{T}_{\mathrm{max}} + \mathrm{T}_{\mathrm{min}})\ /\ 2$
- $T_{\mathrm{base}}$ = base temperature (0°C or 3°C depending on crop stage)
- Cumulative: $\mathrm{GDD}_{\mathrm{total}} = \sum \mathrm{GDD}_{\mathrm{daily}}$

### 6.2.2 Why GDD Instead of Calendar Days?

Calendar-based timing fails across climates:

- Mediterranean climate: Cool spring (slow development)
- Subtropical climate: Warm spring (fast development)
- Same calendar date $\neq$ same crop stage across locations

GDD standardizes: Same GDD accumulation $\approx$ same physiological development.

## 6.3 Zadok Stages

### 6.3.1 Standardized Scale (0–90+)

Figure 6.1 illustrates the progression of wheat through key Zadok stages from sowing to maturity.

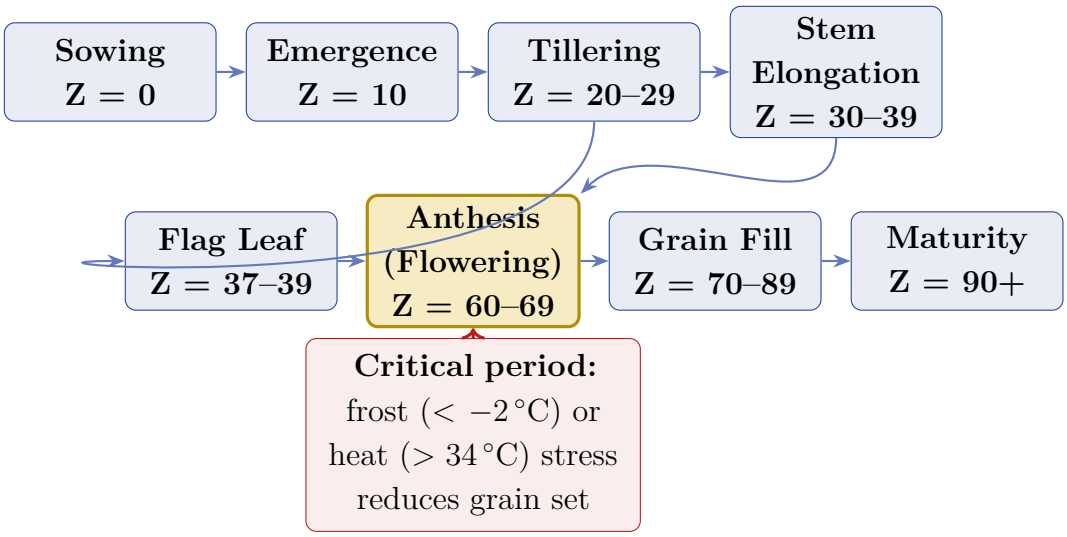


**Figure 6.1:** Simplified wheat phenology on the Zadok scale. The anthesis period (Z60–69, shown in gold) is the most sensitive to temperature extremes. APSIM reports the decimal Zadok stage as `[Wheat].Phenology.Zadok.Stage`.

**Table 6.1:** Zadok Stage Groups and Thermal Time (wheat, subtropical conditions)

| Zadok Stage | Description | GDD (cumulative) | Days after sowing |
|---|---|---|---|
| 0–9 | Germination / Imbibition | 0–50 | 0–10 |
| 10–19 | Emergence (seedling leaves) | 50–100 | 10–20 |
| 20–29 | Leaf development, tiller initiation | 100–300 | 20–60 |
| 30–39 | Stem elongation (growing point differentiates) | 300–600 | 60–120 |
| 40–49 | Booting (flag leaf visible, head in boot) | 600–750 | 120–150 |
| 50–59 | Heading / head emergence | 750–900 | 150–180 |
| 60–69 | Anthesis / pollination | 850–950 | 170–195 |
| 70–79 | Grain development (milk to soft dough) | 950–1100 | 195–240 |
| 80–89 | Grain ripening (hard dough to hard grain) | 1100–1200 | 240–280 |
| 90+ | Physiological maturity / harvest-ready | 1200–1300+ | 280–310 |

**Check against your simulation:** In your APSIM output, the column `[Wheat].Stage` reports Zadok stage as a decimal (e.g., 45.3). Use this to verify your crop is developing at the expected rate.

## 6.4 Critical Periods

Not all growth stages are equally sensitive to stress. Yield loss depends strongly on *when* stress occurs:

| Stage | Zadok | Stress type | Effect on yield |
|---|---|---|---|
| Emergence | 0–10 | Water (dry soil) | Crop fails to establish; zero yield |
| Tillering | 20–39 | N, water | Fewer tillers; fewer potential heads |
| Booting | 40–49 | N, water | Reduced floret number; irreversible |
| Anthesis | 55–65 | Water, heat | Pollen sterility; large grain number loss |
| Grain fill | 70–85 | Water, heat | Smaller grain; 30–50% yield loss if severe |
| Maturity | 90+ | Rain (excess) | Grain sprouting in head; quality loss |

**Anthesis is the most sensitive single stage.** A hot dry day ($T_{max} > 32°C$, ESW declining) at Stage 60–65 can cause widespread pollen sterility and reduce final grain number by 20–50%, regardless of how good the rest of the season is. In APSIM, this is captured through heat and water stress multipliers applied to grain set.

*[IMAGE: Bar chart showing relative yield sensitivity to water stress at each Zadok stage.*
*X-axis: Zadok stage groups. Y-axis: Relative yield impact (%). Anthesis bar is tallest.]*

**Figure 6.2:** Relative sensitivity of wheat yield to water stress at different Zadok stages. Anthesis and early grain fill are the highest-risk periods.

**Why Flowering Date Matters:** Anthesis (Z60–69) is the most temperature-sensitive stage in wheat development. In the Australian grain belt:

- **Frost risk** (night temperature $< -2\,°C$): damages pollen and ovules, causing floret sterility and catastrophic yield loss. Frost risk is highest May–September.
- **Heat stress** (daily max $> 34\,°C$ during flowering): reduces pollen viability and grain set. Risk increases from October onwards in most regions.

The agronomic goal is to position flowering *between* these two risks. Sowing date is the primary management lever used to achieve this.

## 6.5 Sowing Date Effects

### 6.5.1 Why Sowing Date Changes Everything

Shifting the sowing date by 4–8 weeks shifts the entire crop calendar. Crucially, it shifts *which temperatures* the crop experiences at each critical stage:

- **Early May sowing** (Northam, WA): Anthesis falls in August–September — cool, lower heat stress risk; but risk of frost risk at booting (June–July)
- **Mid-June sowing**: Anthesis falls in September–early October — mild temperatures; lowest overall risk; typically optimal for WA Wheatbelt

- **Late July sowing**: Anthesis falls in late October – November — warming; higher heat stress risk; shorter grain fill before hot, dry spring

*[IMAGE: Timeline diagram showing three sowing scenarios. Each is a horizontal bar from Sowing to Maturity.*
*Annotate: anthesis date, frost risk window (Jun–Jul), heat risk window (Oct–Nov) for each scenario.]*

**Figure 6.3:** Effect of sowing date on crop calendar and risk windows. Shifting sowing by 4 weeks moves anthesis into a different temperature regime.

### 6.5.2 Exercise C: Sowing Date Factorial

Three sowing date variants:

- Early (May 1–30): Cool spring; long season
- Mid (June 15–July 15): Moderate; standard
- Late (August 1–31): Warm spring; short season

Expected outcomes:

- Early sowing: Higher tiller number, grain number; risk of frost at anthesis
- Mid sowing: Balanced phenology; maximum grain fill
- Late sowing: Short season; less grain per head; risk of terminal drought

## 6.6 Temperature Stress

### 6.6.1 Heat Stress ($T > 30$°C)

Heat accelerates GDD accumulation (more degree-days per day) but can also damage pollen at anthesis:

- Each extra 1°C above $T_{base}$ adds 1 extra GDD per day
- At $T_{max} > 32$°C during anthesis, pollen viability drops sharply

- Result: Crop matures faster but may set fewer grains

### 6.6.2 Cold Stress ($T < T_{\text{base}}$)

On days where $T_{avg} < T_{\text{base}}$ (0–3°C for wheat), GDD does not accumulate:

- Development pauses; the crop's timeline extends
- If $T_{min} < -2$°C during booting or heading, frost can damage florets irreversibly
- In APSIM, frost damage is applied if temperature falls below a crop-specific threshold during vulnerable stages

### 6.6.3 Vernalization (Winter Wheat)

Some wheat varieties require a cold period (vernalization) to trigger flowering. APSIM tracks a vernalization state: the crop will not advance to reproductive stages until sufficient chilling has accumulated. This is primarily relevant for winter wheat in cooler climates; spring wheat varieties (common in subtropical Queensland) have no or minimal vernalization requirement.

**WA Wheatbelt context:** In the Mediterranean Western Australian Wheatbelt, frost risk is lower than in Queensland but heat and drought stress increase in spring (October–November). An early May sowing means booting occurs in June–July, creating minimal frost risk but extending exposure to spring heat and wind. A mid-June sowing balances these risks by positioning anthesis in September when temperatures are mild.

## 6.7 Connecting Phenology to Management

Understanding phenology helps time key management actions:

| Management action | Target Zadok stage | Reason |
|---|---|---|
| Sowing | Stage 0 | Starting point; everything follows from here |
| N fertiliser application | Stage 30–40 | N demand rising; soil moisture often present |
| Late N (topdress) | Stage 40–50 | Just before booting; directly supports grain protein |
| Insecticide/fungicide | Stage 50–65 | Head/flag leaf protection; avoid anthesis disruption |
| Harvest decision | Stage 90–92 | Grain at maximum dry weight; avoid over-drying |

**In APSIM, the Manager component applies N at sowing** (Stage 0). In reality, split applications often give better results. For this tutorial, we use a single sowing-time application for simplicity.

## 6.8 Summary

**Module 5 key points:**

1. GDD = $\max(0, T_{avg} - T_{base})$ accumulated daily; drives all phenological development
2. Zadok scale (0–90+) is the universal language for describing wheat stages
3. Key milestones: Emergence (Stage 10, ∼50 GDD), Anthesis (Stage 60, ∼900 GDD), Maturity (Stage 90, ∼1200 GDD)
4. Anthesis is the most stress-sensitive stage; N and water stress here cause the largest yield losses
5. Sowing date shifts the entire crop calendar: early sowing risks frost at booting; late sowing risks heat at anthesis
6. Time N applications and management actions using Zadok stage, not calendar date

## 6.9 What's Next?

Module 6 explores long-term climate variability and risk assessment.

# Chapter 7

# Module 6: Long-Term Variability

## 7.1 Learning Objectives

By the end of this module, you will:

- Analyze yield distribution over 20–50 year climate records
- Calculate risk metrics (coefficient of variation, probability of failure)
- Understand the role of rainfall variability in yield
- Design robust management strategies (that work in most years)
- Use multi-year simulations to assess profit variability

## 7.2 Rainfall Variability

### 7.2.1 Inter-Annual Variability in the Western Australian Wheatbelt

Northam (Western Australian Wheatbelt) experiences high year-to-year rainfall variability:

- **Annual rainfall range**: Roughly 300–450 mm/year (long-term mean ≈ 370 mm)
- **Coefficient of Variation (CV) of rainfall**: 30–40% — meaning a "typical" year can be 100–150 mm above or below average
- **ENSO influence**: El Niño years tend to be drier; La Niña years wetter
- **In-season variability**: Drought can occur from October onwards, particularly during critical grain fill period (October–November)

### 7.2.2 Why This Matters for Farming Decisions

If a farmer optimises their management for the average year, they will:

- Over-invest in N in drought years (money wasted; yield limited by water, not N)

- Under-invest in N in wet years (yield potential not reached)

The solution is *risk-aware* management: choose strategies that are profitable across a range of seasons, not just the average.

*[IMAGE: Bar chart of annual rainfall at Northam, 1950–1969 from the weather file.*
*Annotate: long-term mean as horizontal dashed line; highlight 3 driest and 3 wettest years.]*

**Figure 7.1:** Annual rainfall variability at Northam (WA Wheatbelt), 1950–1969 (from Exercise D weather data). Note the high year-to-year variation and lower overall rainfall compared to eastern Australia.

## 7.3 Yield Variability

### 7.3.1 The APSIM Experiment Node

APSIM's **Experiment** node allows you to run multiple treatment combinations automatically. Inside an Experiment, a **Factors** node contains one or more **Factor** nodes that specify which parameter to vary and what values to test.

The Factor specification syntax is:

```
[ComponentName].Script.ParameterName = value1, value2, value3
```

or as a range:

```
[ComponentName].Script.ParameterName = start to end step increment
```

APSIM automatically generates a separate simulation for each combination. For multi-factor experiments, all combinations are created (e.g., 3 sowing dates × 4 N rates = 12 simulations).

### 7.3.2 Exercise D: 20-Year Variability Analysis

Run simulation using 20 consecutive years from climate record (e.g., 1950–1969).

Expected outcomes:

- Wet years: Good autumn/winter rains (May–September) → high ESW at sowing → high yield
- Dry years: Poor in-season rains or hot spring winds → water stress during grain fill (October–November) → low yield
- Yield CV: 30–50% is typical for rainfed wheat in the Mediterranean WA Wheatbelt

*[IMAGE: Time series bar chart of annual wheat yield 1950–1969 from Exercise D.*
*Annotate: mean yield as dashed horizontal line; label the 3 lowest-yield years.]*

**Figure 7.2:** Annual wheat yield variability from the 20-year simulation (Exercise D). The dashed line shows the mean yield.

## 7.4 Risk Metrics

### 7.4.1 Coefficient of Variation

$$\text{CV} = \frac{\text{Std Dev(Yield)}}{\text{Mean(Yield)}}$$

Higher CV = greater uncertainty; farmers prefer lower CV.

### 7.4.2 Probability of Failure

$$P(\text{Yield} < Y_{\text{threshold}}) = \frac{\text{Number of years with Yield} < Y_{\text{threshold}}}{n \text{ total years}}$$

Example: If 5 out of 20 years had yield < 1500 kg/ha, then P(failure) = 5/20 = 25%.

### 7.4.3 Exceedance Probability (Cumulative Distribution)

An exceedance curve shows the probability of exceeding a given yield value. It is constructed by:

1. Sorting all 20 annual yields from lowest to highest
2. Calculating rank: each yield gets a rank $r$ from 1 (lowest) to 20 (highest)

3. Exceedance probability $= (n - r + 1)/n \times 100$

**Reading the exceedance curve:**

- **P50** (50% exceedance): The yield exceeded in 50% of years — the median. A better measure of typical performance than the mean.
- **P80** (80% exceedance): The yield exceeded in 80% of years — the "reliable" yield.
- **P20** (20% exceedance): The yield exceeded only 20% of years — the high-yield potential.

When advising farmers, P80 is the most practically useful: "In 4 out of 5 years, you can expect at least X kg/ha."

## 7.5 Robust Decision Rules

A *robust* management strategy is one that performs reasonably well across the full range of likely seasons — not just the average or best case.

### 7.5.1 Example: N Rate Decision

| N rate (kg/ha) | Mean profit | Worst-year profit | CV (profit) | P(loss) |
|---|---|---|---|---|
| 0 | $540 | $120 | 45% | 5% |
| 80 | $730 | $200 | 40% | 10% |
| 160 | $880 | $180 | 48% | 15% |
| 240 | $850 | $30 | 60% | 25% |

(Indicative values; your Exercise D results will vary.)

- 160 kg/ha maximises mean profit but has higher variability
- 80 kg/ha offers lower mean profit but more reliable income (lower CV)
- 240 kg/ha has the worst risk profile (high N cost, low payoff in dry years)

**The risk aversion principle:** For subsistence or small-scale farmers, minimising catastrophic loss (P(loss)) is often more important than maximising mean profit. Use APSIM multi-year simulations to reveal these trade-offs explicitly before recommending a management strategy.

## 7.6 Climate Change and Yield Risk

A brief introduction (expanded in Module 7):

- If mean temperature rises by 1–2°C, GDD accumulation speeds up — crop matures faster, potentially shortening grain fill
- If rainfall declines (likely in some regions under SSP projections), yield CV will increase and P(failure) rises
- APSIM allows direct testing: modify the weather file (increase $T_{max}$, decrease rainfall) and re-run 20-year simulations to quantify impact

**Climate change increases variability, not just means.** Even if average rainfall stays the same, more extreme droughts and floods can increase yield CV and push P(failure) higher. This is why long-term simulations are essential — single-season models miss this risk.

## 7.7 Summary

**Module 6 key points:**

1. Northern WA Wheatbelt rainfall CV is 30–40%; single-season analysis is insufficient for reliable management recommendations
2. Run 20+ year simulations to capture the full range of climate outcomes
3. Key risk metrics: CV (relative variability), P(failure) = probability yield falls below threshold, P50/P80 from exceedance curve
4. Robust strategies have lower CV and P(failure), even if mean profit is slightly lower
5. Climate change increases variability in addition to shifting mean conditions

## 7.8 What's Next?

Module 7 concludes with climate scenarios and future outlook.

# Chapter 8

# Module 7: Climate Scenarios & Future Outlook

## 8.1 Learning Objectives

By the end of this module, you will:

- Understand how to construct future climate scenarios
- Apply APSIM under baseline vs. climate change weather
- Identify crop varieties and management options that maintain yield
- Assess adaptation feasibility and economics
- Communicate uncertainty to farmers/policymakers

## 8.2 Climate Change and Agriculture

### 8.2.1 Global Context

- Temperature rising: 1.1°C above pre-industrial (as of 2024)
- Rainfall patterns shifting: Some regions wetter, others drier
- Extreme events increasing: Droughts, floods, heat waves
- Crop modeling crucial for adaptation planning

## 8.3 Constructing Climate Scenarios

### 8.3.1 GCM (Global Climate Model) Outputs

Climate models (e.g., IPCC scenarios) provide future temperature and rainfall projections:

- **RCP / SSP scenarios:** Different emission pathways (low, medium, high)
- **Spatial resolution:** Typically 1–50 km grid cells
- **Uncertainty:** Multiple GCMs, ensemble averages

### 8.3.2 Downscaling to Farm Scale

Global Climate Models (GCMs) operate at coarse spatial resolution (50–200 km grids). Before using GCM outputs in APSIM, projections must be *downscaled* to the local scale. Common approaches:

1. **Delta method (simplest)**: Add a monthly temperature offset and multiply rainfall by a scaling factor derived from GCM projections.
   - Example: GCM predicts +1.5°C warming by 2050 → add 1.5 to all $T_{max}$ and $T_{min}$ values in the historical weather file
   - For rainfall: GCM predicts −10% winter rainfall → multiply monthly rainfall by 0.90
2. **Bias correction**: Statistical adjustment so the modelled baseline matches observed historical data (removes systematic GCM biases)
3. **Statistical downscaling**: Relate large-scale atmospheric patterns to local precipitation using regression or weather generators

**For this tutorial**, if you construct a climate scenario, use the delta method: it is transparent, easy to interpret, and sufficient for exploring sensitivity. Open the `.met` weather file in a text editor and modify the temperature columns.

## 8.4 Adaptation Strategies

### 8.4.1 Genetic (Variety Selection)

- Shift to heat-tolerant varieties
- Shift to drought-tolerant varieties (deep roots, higher transpiration efficiency)
- Shift to earlier-maturing varieties (escape late-season drought)

### 8.4.2 Management (Practice Changes)

- Earlier sowing (cooler temperatures pre-anthesis)

- Increased N to maintain protein in warmer years
- Soil water conservation (mulch, reduced tillage)
- Diversification (intercropping, rotation)

### 8.4.3 Infrastructure (Structural Adaptation)

- Irrigation (if water available)
- Soil improvement (increase water-holding capacity)
- Crop insurance (financial risk transfer)

## 8.5 APSIM-Based Adaptation Assessment

### 8.5.1 Step-by-Step: Constructing a Climate Scenario in APSIM

1. **Establish a baseline**: Run your 20-year simulation with historical weather (already done in Exercise D). Record mean yield, CV, and P(failure).
2. **Create a modified weather file**: Make a copy of the `.met` file. Apply the delta method:
   - Add +1.5°C to all $T_{max}$ and $T_{min}$ values (approximate 2050 projection for subtropical QLD)
   - Reduce in-season (May–September) rainfall by 10%
3. **Update the Weather component**: In APSIM, click the `Weather` component → change `FileName` to your modified weather file.
4. **Re-run the simulation**: Keep all other settings identical.
5. **Compare results**: In Excel, place baseline and scenario yields side by side. Calculate:
   - Change in mean yield (kg/ha and %)
   - Change in yield CV
   - Change in P(failure)

*[IMAGE: Side-by-side bar chart comparing annual yields under Baseline vs. +1.5°C scenario.*
*Annotate: mean yield for each scenario as a horizontal dashed line; label years where scenario is notably lower.]*

**Figure 8.1:** Comparison of simulated wheat yields under historical baseline and a +1.5°C warming scenario. Increased temperature accelerates development and shortens grain fill duration.

### 8.5.2 Illustrative Results

Indicative outcomes for a +1.5°C / −10% rainfall scenario at Northam (WA Wheatbelt):

| Metric | Baseline | +1.5°C / −10% rain | Change |
|---|---|---|---|
| Mean yield (kg/ha) | 2800 | 2350 | −16% |
| Yield CV (%) | 28 | 36 | +8 points |
| P(yield < 1500 kg/ha) | 15% | 28% | +13 points |
| Season length (days) | 195 | 180 | −15 days |

(Indicative values for illustration only.)

**Two effects of warming in APSIM:**

1. **Shorter season**: More GDD/day means the crop reaches maturity faster, reducing grain fill duration
2. **Increased water demand**: Higher temperature raises PET, depleting ESW faster during grain fill

Both effects reduce yield in most scenarios. Adaptation strategies target both.

## 8.6 Communicating Uncertainty

**Challenge:** APSIM forecasts have multiple sources of uncertainty:

- Climate model uncertainty (different GCMs give different projections)
- Downscaling uncertainty (statistical method choices)
- Crop model parameter uncertainty (calibration, regional differences)
- Natural variability (internal climate noise; chaotic system)

*Solution:* Present ranges (percentiles), not point estimates. Use ensemble approaches (multiple models, multiple years).

## 8.7 Decision Frameworks

### 8.7.1 Robust Decision Making

Choose strategies that perform reasonably well across multiple scenarios (not optimized for one scenario only).

Example: N rate of 120 kg/ha may not be optimal in any single year, but performs well across dry, normal, and wet years.

## 8.8 Sustainability Considerations

Climate adaptation must also be sustainable:

- **N leaching**: Higher rainfall intensity under some climate scenarios increases drainage and N leaching. APSIM's $NO_3^-$ output tracks this; excess leaching is both an economic loss and an environmental risk.
- **Water use**: Irrigation as an adaptation strategy requires water availability, which may also be declining in some regions.
- **Soil carbon**: Management changes (tillage, residue) affect soil organic matter, which feeds back to mineralisation rates and water-holding capacity — both captured in APSIM's SoilOrganicMatter component.
- **Social acceptance**: Variety changes and new practices require knowledge transfer, seed supply chains, and economic support for adoption.

**Triple bottom line:** The best adaptation strategies improve or maintain yield (economic), reduce environmental impact (ecological), and are feasible for farmers to adopt (social). Use APSIM to assess the first two; consult socioeconomic literature for the third.

## 8.9 Summary

**Module 7 key points:**

1. GCM outputs must be downscaled to farm scale before use in APSIM; the delta method (add T offset, multiply rainfall) is the simplest approach
2. Warming shortens season length (faster GDD accumulation) and increases water demand (higher PET)
3. Multiple RCP/SSP scenarios produce a range of projections — present as ranges, not single values
4. Adaptation strategies: genetic (heat-tolerant varieties), management (earlier sowing, split N), infrastructure (irrigation)
5. Robust decision making: choose strategies that perform acceptably across multiple scenarios
6. Climate adaptation must also be sustainable — consider N leaching, water use, and social feasibility

## 8.10 Synthesis: APSIM in Your Career

This tutorial has covered:

- Simulation mechanics (Module 1)
- Hands-on practice (Modules 2–3, Exercises A–D)
- Agronomic applications (Modules 4–5)
- Risk and adaptation (Modules 6–7)

APSIM is now a tool in your toolkit for:

- Research questions (hypothesis testing via simulation)
- Extension advice (quantifying management options)

- Operational decisions (sowing timing, N optimization)
- Policy analysis (climate impact, food security)

## 8.11 Continuing Your Learning

- APSIM documentation: https://www.apsim.info
- Community forum: APSIM Next Gen GitHub discussions
- Advanced training: APSIM workshops (CIMMYT, ICRISAT, regional centers)
- Job opportunities: APSIM skills highly valued in ag tech, research, extension

# Integrated Exercises

# Appendix A

# Exercise A: Fallow Water Balance

## A.1 Overview

**Objective:** Analyze 7 years of fallow (no crop) simulation to understand water dynamics and identify sowing opportunities.

**File:** `Exercise_A_Fallow_Water_Balance.apsimx`

**Time:** 90 minutes

**Deliverable:** Excel workbook with analysis, plots, and written interpretation.

## A.2 Simulation Setup

### A.2.1 Configuration

**Table A.1:** Exercise A Simulation Parameters

| Parameter | Value |
|---|---|
| Location | Northam, Western Australia (WA Wheatbelt) |
| Climate | 1950–1957 (7-year subset of 101-year record) |
| Weather file | AU_Northam.met |
| Soil | Norwin Vertosol (Black Clay) |
| Crop | None (fallow) |
| Management | None |
| Reported variables | Clock.Today, Soil.ESW, Weather.Rainfall, Soil.Evaporation, Soil.Drainage |

### A.2.2 Key Parameters

**ESW threshold** 100 mm (typical sowing readiness limit)

**Rainfall threshold** 25 mm in last 7 days (recent moisture indicator)

**Sowing window** May 1 – July 10 (typical Queensland wheat season)

## A.3 Task 1: Run Simulation

1. Open APSIM
2. File → Open → `Exercise_A_Fallow_Water_Balance.apsimx`
3. Verify Clock is set to 1950-01-01 to 1957-12-31 (7 years)
4. Click Simulation in tree
5. Press Run (or Ctrl+R)
6. Wait for completion (should be fast; $< 10$ seconds)
7. Right-click DataStore → Export → To Excel
8. Save as `Exercise_A_Results.xlsx`

## A.4 Task 2: Data Exploration

### A.4.1 Examination Checklist

- How many rows in the output? (Should be $\approx 2555$ daily records for 7 years)
- What are the column names? (Should match Report variables: Date, ESW, Rainfall, Evaporation, Drainage)
- Any missing values? (Should be none; every day should have data)
- Date range correct? (1950-01-01 to 1957-12-31)

### A.4.2 Quick Statistics

In Excel, calculate:

- Average ESW across all days
- Maximum ESW per year
- Minimum ESW per year
- Total annual rainfall by year

## A.5 Task 3: Identify Sowing Opportunities

### A.5.1 Criteria

Sowing is possible when BOTH conditions are met:

- ESW > 100 mm
- Rainfall in last 7 days > 25 mm
- Date between May 1 and July 10

### A.5.2 Excel Implementation

1. Create helper columns:
    - Column F: 7-day cumulative rainfall (sum of Rainfall over current + previous 6 days)
    - Column G: Date in sowing window? (IF(AND(Date≥May1, Date≤July10), 1, 0))
    - Column H: ESW sufficient? (IF(ESW>100, 1, 0))
    - Column I: Rainfall sufficient? (IF(F>25, 1, 0))
    - Column J: Sowing ready? (IF(AND(G, H, I), 1, 0))
2. Count: How many days are sowing-ready? (COUNTIF(J, 1))

## A.6 Task 4: Visualization

### A.6.1 Plot 1: ESW Time Series

- X-axis: Date (all 7 years)
- Y-axis: ESW (mm)
- Add reference line: ESW = 100 mm (sowing threshold)
- Title: "Extractable Soil Water: 7-Year Fallow"

**Interpretation:** When does ESW peak? When is it lowest? Pattern seasonality?

### A.6.2 Plot 2: Annual ESW Pattern

- Calculate monthly average ESW (group by year-month)
- Plot 7 overlaid lines, one per year (different colors)
- X-axis: Month (1–12)
- Y-axis: Average ESW (mm)
- Title: "Monthly ESW Pattern: 7 Years"

**Interpretation:** Is seasonal pattern consistent across years? When does ESW typically peak?

### A.6.3 Plot 3: Rainfall vs. ESW Change

- Create column: Daily ESW change = ESW(today) - ESW(yesterday)
- Scatter plot: X = Daily Rainfall, Y = Daily ESW change
- Add trend line
- Title: "Daily Rainfall vs. ESW Change"

**Interpretation:** How much does ESW change per mm of rainfall? Any threshold effects?

## A.7 Task 5: Analysis Questions

Answer the following using your data:

1. **Seasonal Pattern:** ESW is typically highest in [season?] and lowest in [season?]. Why?
2. **Sowing Ready:** In how many days per year (on average) are conditions suitable for sowing (ESW > 100 mm, rain threshold, date window)? What is the range across the 7 years?
3. **Dry Year vs. Wet Year:** Identify the driest and wettest year in your data (by total rainfall). Compare their ESW profiles. What are the implications for management?
4. **Water Loss Processes:** For a typical year, calculate total water loss to:
   - Evaporation (sum of daily Evaporation values)
   - Drainage (sum of daily Drainage values)

   Which process dominates? How does it vary seasonally?

5. **Sowing Decision Rule:** Propose a sowing rule for farmers in this region. Should they wait for ESW > 100 mm? What rainfall accumulation period is most predictive (7 days? 14 days)? What date window makes sense?

## A.8 Task 6: Comparison Scenario (Extension)

*Optional:* Modify the simulation to explore:

### A.8.1 Scenario A: Shallow Soil

Create a variant with half the soil depth (lower water-holding capacity). How does ESW pattern change? Sowing opportunities more or less frequent?

### A.8.2 Scenario B: Different Climate

Substitute a wetter region's weather file (or a drier region). Repeat analysis. Compare.

## A.9 Deliverable

Submit a single `Exercise_A_Report.xlsx` or `Exercise_A_Report.docx` containing:

- ✓ Analyzed data (with helper columns for sowing readiness)
- ✓ 3+ visualizations (plots as described above)
- ✓ Written answers to Analysis Questions (5 items)
- ✓ Brief reflection (100–150 words): What surprised you? How would you use this in a real farming decision?

## A.10 Grading Rubric

| Criterion | Points | Description |
|---|---|---|
| Simulation setup | 10 | File opened, run successfully, data exported |
| Data exploration | 10 | Correct record count, column names, date range verified |
| Sowing identification | 15 | Correct criteria applied; counts accurate |
| Visualization (3 plots) | 30 | Clear labels, interpretable scales, titles |
| Analysis answers | 25 | Thoughtful interpretation of data; connections to concepts |
| Reflection | 10 | Personalized insights; practical relevance |
| **Total** | **100** | |

# Appendix B

# Exercise B: Nitrogen Response Factorial

## B.1 Overview

**Objective:** Run four N rate treatments and analyze yield response to determine optimal N rate and calculate agronomic efficiency.

**Files:**

- `Exercise_B_N0.apsimx` (0 kg/ha N)
- `Exercise_B_N80.apsimx` (80 kg/ha N)
- `Exercise_B_N160.apsimx` (160 kg/ha N)
- `Exercise_B_N240.apsimx` (240 kg/ha N)

**Time:** 90 minutes (run all 4 simulations; 10 min per run)

**Deliverable:** Excel workbook with N response curve, NUE calculations, cost-benefit analysis.

## B.2 Simulation Setup

### B.2.1 Configuration

**Table B.1:** Exercise B Simulation Parameters

| Parameter | Value |
|---|---|
| Location | Northam, Western Australia |
| Climate | 1949–1950 (single growing season) |
| Weather file | AU_Northam.met |
| Soil | Norwin Vertosol |
| Crop | Wheat (Hartog cultivar) |
| Sowing date | May 15 (variable rule: ESW > 100 mm + rain threshold) |
| Fertiliser rate | 0, 80, 160, 240 kg/ha $NO_3$-N (applied at sowing) |
| Reported variables | Clock.Today, Wheat.LAI, Wheat.Phenology.Zadok.Stage, Wheat.Grain.Total.Wt, Wheat.Grain.Protein, Wheat.AboveGround.N |

## B.3 Task 1: Run All Four Simulations

1. Open Exercise_B_N0.apsimx; run (export to `Exercise_B_N0_Results.xlsx`)
2. Open Exercise_B_N80.apsimx; run (export to `Exercise_B_N80_Results.xlsx`)
3. Open Exercise_B_N160.apsimx; run (export to `Exercise_B_N160_Results.xlsx`)
4. Open Exercise_B_N240.apsimx; run (export to `Exercise_B_N240_Results.xlsx`)

   *Note:* Each simulation should take < 5 minutes

## B.4 Task 2: Extract Grain Yield

### B.4.1 Approach

- For each .xlsx file, find the row with maximum Wheat.Grain.Total.Wt
- This represents harvest time (physiological maturity)
- Record as final yield for that N rate
- Create summary table: N rate vs. Final Yield

**Table B.2:** Expected N Response Pattern

| N Rate (kg/ha) | Yield (kg/ha) | NUE (kg/kg) |
|---|---|---|
| 0 | 2000 | – |
| 80 | 3500 | (3500–2000)/80 = 18.75 |
| 160 | 4600 | (4600–2000)/160 = 16.25 |
| 240 | 5200 | (5200–2000)/240 = 13.33 |

*(Exact numbers will vary; this is illustrative)*

## B.5 Task 3: Calculate Nitrogen Use Efficiency (NUE)

$$\text{NUE}_i = \frac{\text{Yield}_i - \text{Yield}_{0\ \text{kg/ha}}}{N\ \text{rate}_i} \quad [\text{kg grain / kg N}]$$

Example calculation:

- Yield(N=0) = 2000 kg/ha
- Yield(N=80) = 3500 kg/ha
- NUE(80) = (3500 – 2000) / 80 = 18.75 kg grain / kg N

**Observations:**

- NUE is highest at low N rates (diminishing returns)
- NUE decreases as N rate increases (approaching physiological limit)
- NUE reflects N use efficiency; typical wheat: 10–20 kg/kg

## B.6 Task 4: Plot N Response Curve

- X-axis: N rate (0, 80, 160, 240 kg/ha)
- Y-axis: Yield (kg/ha)
- Plot points for each N rate
- Add polynomial trendline (degree 2 is typical)
- Title: "Nitrogen Response Curve: Wheat"

**Interpretation:** Where is the curve flat (diminishing returns)? At what N rate does yield plateau?

## B.7 Task 5: Cost-Benefit Analysis

### B.7.1 Profit Calculation

$$\text{Profit} = (\text{Yield} \times P_{\text{grain}}) - (\text{N rate} \times P_{\text{N}})$$

### B.7.2 Assumptions

- Grain price: \$400/tonne (adjust based on current market)
- N cost: \$1.50/kg (adjust based on fertiliser price)

### B.7.3 Example Calculation

For N = 160 kg/ha:

- Yield = 4600 kg/ha = 4.6 tonnes/ha
- Revenue = 4.6 × \$400 = \$1840/ha
- N cost = 160 × \$1.50 = \$240/ha
- Profit = \$1840 – \$240 = \$1600/ha

### B.7.4 Create Profit Table

| N (kg/ha) | Yield (t/ha) | Revenue | N Cost | Profit |
|---|---|---|---|---|
| 0 | – | – | \$0 | – |
| 80 | – | – | \$120 | – |
| 160 | – | – | \$240 | – |
| 240 | – | – | \$360 | – |

## B.8 Task 6: Analysis Questions

1. What is the yield response to N? (How much yield increases per kg N?)
2. At what N rate does NUE drop to 10 kg/kg? (Where do diminishing returns become pronounced?)

3. What is the profit-maximizing N rate (assuming prices: $400/tonne grain, $1.50/kg N)?
4. If grain price fell to $300/tonne, would you increase or decrease N rate? Why?
5. If N cost rose to $2.50/kg, how would profit be affected?

## B.9 Task 7: Grain Quality

### B.9.1 Protein Analysis

- Extract Wheat.Grain.Protein from each simulation at harvest
- Plot: N rate vs. Protein content
- Interpretation: Does higher N increase grain protein? By how much?

*Expected finding:* Grain protein increases with N rate (premium prices for high-protein wheat).

## B.10 Deliverable

Submit `Exercise_B_Report.xlsx` or `Exercise_B_Report.docx`:

- ✓ Yield summary table (N rate vs. Yield)
- ✓ NUE calculations (table with NUE for each N rate)
- ✓ N response curve plot
- ✓ Profit analysis table + chart
- ✓ Protein vs. N rate plot
- ✓ Written answers to 7 analysis questions
- ✓ Conclusion: Your recommendation for optimal N rate and justification

# Appendix C

# Exercise C: Sowing Date Factorial

## C.1 Overview

**Objective:** Run three near-solved simulations with different sowing date windows and compare the effects on crop development, yield, and grain quality.

| | **Early May** | **Mid June** | **Late July** |
|---|---|---|---|
| File | `Exercise_C_EarlyMay.apsimx` | `Exercise_C_MidJun.apsimx` | `Exercise_C_Late` |
| Sowing window | 1 May – 31 May | 15 Jun – 15 Jul | 1 Aug – 31 Aug |
| Duration | 90 minutes | | |

**Simulated season:** 1949-07-01 to 1950-12-31 (one full growing season)

**Location:** Northam, Western Australia

**Deliverables:** Summary table, phenology comparison chart, yield table, written analysis

## C.2 Background

Sowing date is one of the most powerful management levers available to a wheat farmer. Because wheat development is driven by thermal time (GDD), shifting the sowing date by 4–6 weeks moves the entire crop calendar, including the most stress-sensitive stages (anthesis, grain fill), into different temperature and rainfall conditions.

**The core question:** For a given location and season, which sowing date best aligns the critical period (booting to grain fill) with favorable temperatures and soil water availability?

## C.3 Task 1: Open and Review the Simulation Files

1. Open `Exercise_C_EarlyMay.apsimx` in APSIM

2. Click the `Manager` component in the simulation tree
3. Locate the sowing rule. You will see two parameters:
   - `StartDate`: "1-may"
   - `EndDate`: "31-may"
4. Note the sowing conditions: ESW > 100 mm AND 7-day rainfall > 25 mm
5. Open `Exercise_C_MidJun.apsimx` and `Exercise_C_LateJul.apsimx`; verify that all other parameters (soil, variety, N rate, weather file) are identical — only the sowing window differs

*[IMAGE: APSIM screenshot showing the Manager component properties panel for one of the Exercise C files.*
*Annotate: StartDate and EndDate parameters highlighted in the sowing rule.]*

**Figure C.1:** Manager component in Exercise C showing the sowing window parameters. Only StartDate and EndDate differ between the three files.

## C.4 Task 2: Run All Three Simulations and Export Results

1. Open `Exercise_C_EarlyMay.apsimx`; press **Run** (F5)
2. Wait for completion (progress bar at bottom)
3. Export results: Results tab → right-click → Export → Excel
4. Save as `C_EarlyMay_Results.xlsx`
5. Repeat for `Exercise_C_MidJun.apsimx` and `Exercise_C_LateJul.apsimx`

**Efficiency tip:** You can run all three simulations in sequence before exporting. APSIM stores results from each run separately in its DataStore.

## C.5 Task 3: Phenological Comparison

### C.5.1 Extract Key Dates

For each simulation, find the date when each Zadok stage is first reached. In your exported Excel file:

1. Add a helper column: `=YEAR(A2)` and `=MONTH(A2)` for date filtering
2. Use `MINIFS` to find the date when `[Wheat].Stage` first crosses each threshold:
   - Emergence: Stage $\geq$ 10 (`=MINIFS($A:$A, $C:$C, ">=10")` where column C = Stage)
   - Booting: Stage $\geq$ 45
   - Anthesis: Stage $\geq$ 60
   - Physiological maturity: Stage $\geq$ 90

### C.5.2 Thermal Time Accumulation

For each scenario, calculate cumulative GDD from sowing date to anthesis:

1. Identify the sowing date (first row where Stage $>$ 0)
2. Identify the anthesis date (first row where Stage $\geq$ 60)
3. In a Python or Excel formula, sum the daily GDD column for that date range

**Expected result:** All three scenarios should accumulate similar GDD from sowing to anthesis ($\sim$850–950 GDD). The differences will be in *how many calendar days* this takes — cool early-winter sowing takes more calendar days to accumulate the same GDD as a warmer late-winter sowing.

## C.6 Task 3: Phenological Comparison

### C.6.1 Key Dates

Track dates for each sowing scenario:

- Sowing date
- Emergence date (Zadok stage 10)

- Booting date (Zadok stage 40–49)
- Anthesis date (Zadok stage 60–69)
- Physiological maturity / Harvest date (Zadok stage 90)

### C.6.2 Thermal Time Accumulation

Calculate cumulative GDD from sowing to anthesis. Compare across sowing dates.

## C.7 Task 4: Yield and Quality Comparison

### C.7.1 Harvest Results Table

For each scenario, extract from the row where Zadok Stage = 90 (harvest):

| **Metric** | **Early May** | **Mid June** | **Late July** |
|---|---:|---:|---:|
| Actual sowing date | *record* | *record* | *record* |
| Emergence date (Stage 10) | *record* | *record* | *record* |
| Anthesis date (Stage 60) | *record* | *record* | *record* |
| Maturity date (Stage 90) | *record* | *record* | *record* |
| Season length (days) | *calculate* | *calculate* | *calculate* |
| GDD sowing to anthesis | *calculate* | *calculate* | *calculate* |
| Grain yield (kg/ha) | *record* | *record* | *record* |
| Biomass at harvest (kg/ha) | *record* | *record* | *record* |
| Harvest Index | *calculate* | *calculate* | *calculate* |
| Grain protein (%) | *record (if available)* | | |

### C.7.2 Expected Sowing Date Effects

**Table C.1:** Expected outcomes by sowing date (indicative)

| Metric | Early May | Mid June | Late July |
|---|---|---|---|
| Season length | Long (∼230+ days) | Moderate (∼195 days) | Short (∼165 days) |
| Calendar days, sowing to anthesis | Many (cool autumn/winter) | Moderate | Few (warm spring) |
| Anthesis temperature | Cool (Aug) | Mild (Sep) | Warm (Oct–Nov) |
| Frost risk at booting | Higher (July booting) | Lower | Very low (Sep booting) |
| Heat stress at anthesis | Very low | Low | Moderate |
| Grain fill duration | Long | Moderate | Short |
| Grain yield | Moderate–high (frost risk) | Typically highest | Lower (short fill) |
| Grain protein | Low–moderate | Moderate | Higher (heat stress) |
| Harvest Index | Moderate–high | High (∼0.45) | Lower (∼0.38) |

## C.8 Task 5: Analysis Questions

Answer the following questions using your harvest results table and plots. Support each answer with specific numbers from your simulation output.

1. **Calendar days vs. thermal time:** Compare the number of calendar days from sowing to anthesis across the three scenarios. Then compare cumulative GDD from sowing to anthesis. Which is more similar across scenarios — calendar days or GDD? What does this tell you about how APSIM models crop development?

2. **Yield response:** Which sowing date produced the highest grain yield in this season? Using the ESW output and $T_{max}$ output, explain *why* that scenario outperformed the others. Was it related to water availability, temperature, or both?

3. **Critical period and stress:** For the Late July sowing, identify the date of anthesis in your output. What was the $T_{max}$ on that date? Was ESW above the 100 mm sowing threshold at that time? Does the output suggest water stress or heat stress contributed to any yield reduction?

4. **Season length and grain fill:** Calculate season length (sowing to maturity) for

each scenario. How does grain fill duration (maturity minus anthesis) change with sowing date? How might a shorter grain fill period explain differences in yield and grain size?

5. **Harvest Index:** Calculate Harvest Index for each scenario (Grain mass / Biomass). Which scenario has the highest HI? What does a lower HI indicate about where assimilate ended up (straw vs. grain)?

6. **Practical recommendation:** Based on your simulation results for this one season (1949–1950), which sowing date would you recommend to a Northam wheat farmer in the WA Wheatbelt? Acknowledge at least one limitation of making this recommendation from a single-season analysis.

## C.9 Task 6: Visualisation

Create the following two plots:

### C.9.1 Crop Development Comparison (Line Plot)

1. In Excel (or Python), plot Zadok Stage (y-axis) vs. Date (x-axis) for all three scenarios on the same axes

2. Use different line colours/styles for each scenario

3. Add vertical dashed lines at: frost risk window (Jul 1 – Jul 31) and heat risk window (Oct 1 – Nov 30)

4. Add horizontal dashed lines at Stage 45 (Booting), Stage 60 (Anthesis), Stage 90 (Maturity)

5. Title: "Phenological Development by Sowing Date"

*[IMAGE: Student-generated Zadok stage vs. date line plot for all three sowing date scenarios.*
*Annotate: frost risk window shaded in blue; heat risk window shaded in red.]*

**Figure C.2:** Example student output: Zadok stage trajectory for three sowing dates. The frost and heat risk windows highlight the trade-off between early and late sowing.

### C.9.2 Yield and Harvest Index Comparison (Bar Chart)

Create a grouped bar chart with:

- X-axis: three sowing date scenarios
- Two bar groups: Grain yield (left axis, kg/ha) and Harvest Index (right axis, 0–1)
- Title: "Grain Yield and Harvest Index by Sowing Date"

## C.10 Deliverable

Submit a single Excel workbook (or Python notebook + PDF) containing:

1. **Harvest results table** (Task 4): all metrics for three scenarios
2. **Phenology comparison chart** (Task 6, Plot 1): Zadok stage vs. date
3. **Yield/HI bar chart** (Task 6, Plot 2): grouped bar chart
4. **Written analysis** (Task 5): six questions, approximately 2–4 sentences each
5. **Recommendation** (Task 5, Q6): one paragraph with supporting evidence

**Total time:** 90 minutes

**Assessment criteria:**

- Harvest table complete and values are plausible (20%)
- Plots are correctly labelled with units and titles (20%)
- Analysis questions answered with specific data values (40%)
- Recommendation is evidence-based and acknowledges limitations (20%)

# Appendix D

# Exercise D: 20-Year Variability Analysis

## D.1 Overview

**Objective:** Run a 20-year continuous simulation and analyze yield distribution, identify dry/wet years, assess management robustness.

**File:** `Exercise_D_20Year_Variability.apsimx`

**Time:** 120 minutes (includes data processing and visualization)

**Deliverable:** Comprehensive analysis of yield variability and risk metrics.

## D.2 Simulation Setup

### D.2.1 Configuration

**Table D.1:** Exercise D Simulation Parameters

| Parameter | Value |
|---|---|
| Location | Northam, Western Australia |
| Climate | 20 consecutive years (e.g., 1950–1969) |
| Weather file | AU_Northam.met |
| Soil | Norwin Vertosol |
| Crop | Wheat (Hartog cultivar) |
| Management | Fixed: Sow May 1–July 10 (ESW > 100 mm); N = 160 kg/ha |
| Report variables | Clock.Today, Wheat.Grain.Total.Wt, Weather.Rainfall, Soil.ESW |

## D.3 Task 1: Run Simulation

1. Open `Exercise_D_20Year_Variability.apsimx`
2. Verify Clock: start 1950-01-01, end 1969-12-31 (20 years)

3. Run simulation ($\sim$ 30 seconds)
4. Export to `Exercise_D_Results.xlsx`
5. Verify: $\approx$ 7300 daily records (365 $\times$ 20 years)

## D.4 Task 2: Extract Annual Yields

### D.4.1 Approach

1. Create helper column: Year (extract from date)
2. Create helper column: Season Year (e.g., May 1, 1950 = year 1950 season; previous May 1 thru Apr 30)
3. For each season year, find maximum Wheat.Grain.Total.Wt (represents harvest)
4. Create summary table: Season Year vs. Final Yield

### D.4.2 Result Example

**Table D.2:** Annual Yield Summary (Illustrative)

| Season Year | Annual Rainfall (mm) | Final Yield (kg/ha) | ESW at Sowing (mm) |
|---|---|---|---|
| 1950 | 450 | 3800 | 125 |
| 1951 | 380 | 2100 | 95 |
| *(wet year)* | *(high)* | *(high)* | – |
| *(dry year)* | *(low)* | *(low)* | – |
| ... | ... | ... | ... |

## D.5 Task 3: Calculate Risk Metrics

### D.5.1 Basic Statistics

- Mean yield: $\bar{Y} = \frac{1}{n}\sum Y_i$
- Std dev: $\sigma = \sqrt{\frac{1}{n-1}\sum(Y_i - \bar{Y})^2}$
- Min/Max: Identify driest and wettest years
- Coefficient of variation: CV $= \sigma/\bar{Y}$

### D.5.2 Risk Metrics

$$P(\text{Yield} < 1500 \text{ kg/ha}) = \frac{\#\text{ years with Yield} < 1500}{20 \text{ years}}$$

*Interpretation:* If P = 0.25, then 1 in 4 years (25%) has yield below 1500 kg/ha; risky threshold.

**Robustness Definition:** A management practice is "robust" if it maintains yield above a threshold (e.g., 2000 kg/ha) in at least 80% of years. High-risk scenarios fail in > 20% of years.

## D.6 Task 4: Visualize Yield Distribution

### D.6.1 Plot 1: Annual Yield Time Series

- X-axis: Year (1950–1969)
- Y-axis: Yield (kg/ha)
- Add horizontal lines: Mean, Mean±1SD (shaded band)
- Title: "Annual Wheat Yield: 20-Year Variability"

### D.6.2 Plot 2: Histogram of Yields

- X-axis: Yield classes (e.g., 1000–1500, 1500–2000, etc.)
- Y-axis: Frequency (number of years)
- Overlay: Normal distribution curve
- Title: "Yield Distribution"

### D.6.3 Plot 3: Rainfall vs. Yield Scatter

- X-axis: Annual rainfall
- Y-axis: Annual yield
- Add regression line
- Title: "Rainfall vs. Yield Relationship"

## D.7 Task 5: Identify Dry and Wet Years

### D.7.1 Classification

Define dry/wet years using rainfall or yield:

- Dry: Yield < (Mean – 1 SD) or Rainfall < 25th percentile
- Average: (Mean – 0.5 SD) to (Mean + 0.5 SD)
- Wet: Yield > (Mean + 1 SD) or Rainfall > 75th percentile

### D.7.2 Analysis

1. Identify specific dry years (e.g., 1952, 1957)
2. Identify specific wet years (e.g., 1955, 1960)
3. Compare phenology, grain number, protein for dry vs. wet years
4. Hypothesize: Why does 1952 fail while 1951 doesn't?

## D.8 Task 6: Analysis Questions

1. What is the average yield? Coefficient of variation? Interpretation in terms of farmer risk?
2. In how many years does yield exceed 3000 kg/ha? (Probability of good harvest)
3. In how many years does yield fall below 1500 kg/ha? (Risk threshold)
4. Which weather variable (rainfall, temperature, solar radiation) most affects yield in this region?
5. If a farmer needs minimum 2000 kg/ha to be profitable, what is their risk of failure? Acceptable?
6. Propose an adaptation: Should farmers shift sowing date (earlier? later?) to reduce risk? Or increase N? Or plant a different variety? Justify using your data.

## D.9 Task 7: Scenario Analysis (Extension)

### D.9.1 Scenario A: Increased N (240 kg/ha)

Re-run simulation with higher N. Does it reduce year-to-year variability? Why or why not?

### D.9.2 Scenario B: Earlier Sowing

Shift sowing window to April–June. How does yield variability change?

### D.9.3 Scenario C: Climate Change

Modify weather data (e.g., +2°C, –10% rainfall). Run 20-year simulation. Compare yield distribution.

## D.10 Deliverable

Submit `Exercise_D_Report.xlsx` or `Exercise_D_Report.docx`:

- ✓ Annual yield summary table (20 years)
- ✓ Risk metrics (mean, SD, CV, P(Yield < threshold))
- ✓ 3+ visualizations (time series, histogram, scatter)
- ✓ Dry vs. wet year comparison (phenology, yields)
- ✓ Written answers to 7 analysis questions
- ✓ Recommendation: Is the current management robust? What adaptation would you propose?

# Appendix E

# APSIM Components Reference

## E.1 Global Components

### E.1.1 Clock

`Clock` advances simulation by 1 day. User specifies:

- Start date (e.g., 1900-01-01)
- End date (e.g., 2000-12-31)
- Timestep (always 1 day in APSIM)

Role: Synchronizes all other components to daily cycle.

### E.1.2 Weather

`Weather` reads external file (`.met` format) with daily:

- MaxT, MinT (°C)
- Rainfall (mm)
- Radiation (MJ/m$^2$/day)
- RelativeHumidity (%)
- WindSpeed (m/s)

Role: Provides atmospheric drivers; calculates PET (Potential Evapotranspiration).

### E.1.3 SoilArbitrator

`SoilArbitrator` coordinates water and nutrient availability among competing processes.

Role: Ensures consistency when multiple components access soil (plant uptake, evaporation, nitrification).

### E.1.4 MicroClimate

`MicroClimate` partitions solar radiation and calculates surface energy balance.

Role: Computes interception, albedo, near-surface temperature; feeds into soil heat flux.

## E.2 Zone Components

### E.2.1 Soil

`Soil` represents multi-layer soil profile (typically 7–10 layers).

**Sub-components**

**`Physical`** Bulk density, water characteristics (LL, DUL, SAT), saturated conductivity for each layer; includes crop-specific (`SoilCrop`) parameters (LL, KL, XF)

**`SoilWater`** (WaterBalance model) Calculates daily water budget (infiltration, drainage, evaporation, plant uptake); tracks water stress factors

**`Organic`** Organic matter pools (Fresh Organic Matter, Humic Pool) and C:N ratios; mineralization kinetics

**`Chemical`** pH, electrical conductivity, exchangeable sodium percentage; available for future expansion (salinity, boron, etc.)

**`Nutrient`** (or `Organic`) Organic N pool; mineralization-immobilization rates

**`Solutes`** Individual solute pools: $NO_3$, $NH_4$, Urea; movement with water; leaching

**`Temperature`** Soil temperature by depth; heat diffusion; important for mineralization, nitrification rates

### E.2.2 Crop (e.g., Wheat)

`Wheat` (or other crop model) simulates plant processes:

**`Phenology`** Thermal time accumulation; Zadok stages; development rate parameters by stage

**`Leaf`** LAI (Leaf Area Index) progression; specific leaf area (SLA); leaf senescence

**`Root`** Root depth progression; water and N extraction coefficients; root biomass

**`Biomass`** Daily photosynthesis (RUE: Radiation Use Efficiency); respiration; senescence; partitioning to organs

**`Grain`** Grain number determination; grain fill duration; harvest index; grain N content

### E.2.3 Manager

`Manager` user-defined control logic (JavaScript or C#).

Typical rules:

- Sowing: IF (date in window AND ESW sufficient AND recent rain) THEN Sow()
- Fertilisation: IF (date or stage trigger) THEN Fertiliser.Apply()
- Harvest: IF (crop physiologically mature) THEN Harvest()

### E.2.4 Report

`Report` specifies output variables (acts as filter).

User lists variables to record each day; APSIM records only selected variables (reduces output file size).

### E.2.5 DataStore

`DataStore` SQL-like database; receives Report data each day; exports to Excel/CSV at simulation end.

## E.3 Variable Naming Convention

APSIM variables use dot notation:

```
[ComponentName].SubComponent.Variable

Examples:
[Wheat].LAI                          # Leaf Area Index
[Wheat].Grain.Total.Wt               # Total grain mass
[Soil].ESW                           # Extractable soil water
[Soil].SoilWater.Evaporation         # Daily evaporation
[Weather].Rainfall                   # Daily rainfall
[Clock].Today                        # Current date
```

**Note:** Component names are enclosed in square brackets when used in Report variable lists.

## E.4 Variable Quick Reference

Table E.1 lists the most commonly used APSIM report variables for wheat simulations in the WA Wheatbelt.

**Table E.1:** Commonly used APSIM Report variables for wheat simulations.

| Variable | Description |
|---|---|
| `[Clock].Today` | Simulation date |
| `[Wheat].Grain.Total.Wt*10 as Yield` | Grain yield (kg/ha) |
| `[Wheat].Grain.Protein` | Grain protein (%) |
| `[Wheat].Phenology.Zadok.Stage` | Zadok growth stage (decimal) |
| `[Wheat].Phenology.CurrentStageName` | Stage name as text |
| `[Wheat].Phenology.DaysAfterSowing` | Days since sowing |
| `[Wheat].LAI` | Leaf area index ($m^2/m^2$) |
| `[Wheat].AboveGround.Wt` | Above-ground biomass ($g/m^2$) |
| `[Wheat].AboveGround.N` | Above-ground N content (g $N/m^2$) |
| `sum([Soil].SoilWater.ESW)` | Total extractable soil water (mm) |
| `[Soil].SoilWater.Runoff` | Daily runoff (mm) |
| `[Soil].SoilWater.Drainage` | Daily drainage below root zone (mm) |
| `[Soil].SoilWater.Es` | Soil surface evaporation (mm) |
| `sum([Soil].NO3.kgha) as NO3Total` | Total soil nitrate (kg N/ha) |
| `sum([Soil].NH4.kgha) as NH4Total` | Total soil ammonium (kg N/ha) |
| `[Weather].Rain` | Daily rainfall (mm) |
| `[Weather].MaxT` | Daily maximum temperature (°C) |

# Appendix F

# Troubleshooting Common Errors

This appendix provides diagnostic decision trees for common APSIM problems.

## F.1 Crop Won't Sow

### F.1.1 Diagnostic Flowchart

1. **Q: Does simulation run without errors?**
   - YES → Go to 2
   - NO → Check file format, component structure; see APSIM documentation
2. **Q: Check final output: Is Zadok stage always 0 (not sown)?**
   - YES → Crop is not sowing; go to 3
   - NO → Crop did sow at some point; issue is different
3. **Q: Is there a Manager rule for sowing?**
   - YES → Go to 4
   - NO → Add Manager rule (see Module 1); crop won't sow without rule
4. **Q: Check Manager parameters: Is ESW threshold too high?**
   - Check: `MinESW` parameter. If MaxESW in simulation < MinESW, crop never meets threshold
   - Solution: Reduce MinESW or increase initial soil water
5. **Q: Check Manager parameters: Is sowing window correct?**
   - Check: `StartDate`, `EndDate`. Are they in correct format (d-mmm)?
   - Example: "1-may" (correct), "May 1" (incorrect)
   - Solution: Use d-mmm format

6. **Q: Is rainfall threshold met?**
   - Check: Weather file data. Is there a 7-day period with rainfall > 25 mm during sowing window?
   - Solution: If region is always dry, reduce rainfall threshold or choose wetter year from weather file

## F.2 Unrealistic Yield

### F.2.1 Yield Too Low

**Possible causes:**

1. Crop sown too late (short season, incomplete grain fill)
2. Insufficient N (N stress throughout season)
3. Water stress (low rainfall or high drainage)
4. Crop encounters frost during booting/anthesis

**Diagnostic:**

- Plot Zadok stage vs. date; check if crop reaches grain fill (stages 70–89)
- Plot [Soil].ESW vs. date; check for critical stress periods (booting, grain fill)
- Check Manager: Is sowing happening at right time?
- Check N rate: Is it sufficient (compare N=0 vs. N=160)?

### F.2.2 Yield Too High

**Possible causes:**

1. Initial conditions unrealistic (soil too wet, too much initial N)
2. Weather file synthetic or biased (unrealistically favorable)
3. Report variables wrong (grain mass not realistic for cultivar/location)

**Diagnostic:**

- Compare simulated yield to local farm data / research records
- Check initial soil water (`InitialPAWmm`) and initial N pools
- Verify weather file source and quality

## F.3 No Output or Empty Report

### F.3.1 DataStore Empty

**Possible causes:**

1. Report component missing or disabled
2. Report variables list is empty
3. Simulation didn't run (errors in setup)

**Diagnostic:**

- In APSIM GUI, click Report in tree; check Properties panel
- Verify: VariableNames array is not empty
- Verify: Report is Enabled (checkbox ticked)
- Re-run simulation; check output window for errors

### F.3.2 Report Variables Have Missing Values

**Possible causes:**

1. Variable path is incorrect (typo in [Component].Variable naming)
2. Component not initialized when Report tries to access it (e.g., [Wheat].LAI before crop sown)
3. Component is disabled

**Diagnostic:**

- Verify variable names match APSIM conventions (**??**)
- Use APSIM GUI to auto-generate variable names (reduces typos)
- Check if missing values coincide with off-season (e.g., wheat LAI=0 when crop not alive)

## F.4 File Path Errors

**Error message:** "Cannot find weather file AU_Northam.met"

**Diagnosis:**

- Check Weather component: FileName property uses `%root%` variable
- `%root%` = APSIM installation directory
- Example: `C:/Program Files/APSIM2025/Examples/WeatherFiles/AU_Northam.met`

**Solution:**

- Verify weather file exists in APSIM Examples folder
- If not, download from APSIM repository or use a different weather file
- Alternatively, use absolute path (if sharing files, portability is lost)

## F.5 JSON Parsing Errors

**Error message:** "Error parsing .apsimx file: JSON format invalid"

**Cause:** Manually edited .apsimx file has JSON syntax error (mismatched braces, quotes, etc.)

**Solution:**

- Use JSON validator (paste file contents at jsonlint.com)
- Use APSIM GUI to edit instead of text editor
- If editing manually, use code editor with JSON highlighting (VS Code, Sublime, etc.)

## F.6 Simulation Hangs or Crashes

**Symptom:** Simulation starts but doesn't complete; runs for hours.

**Possible causes:**

1. Very long time period (e.g., 1000 years) or complex soil model
2. Infinite loop in Manager code
3. Numerical instability (rare)

**Solution:**

- Start with short simulations (1–5 years) to test setup
- Check Manager JavaScript for logical errors
- If simulation is legitimately long, be patient or increase hardware

## F.7 Getting Help

If you encounter an error not listed above:

1. Check **APSIM Documentation**: https://www.apsim.info
2. Post on **APSIM Forum**: https://github.com/APSIMInitiative/ApsimX/discussions
3. Consult **Wiki Resources**: `FAQ_Crop_Won't_Emerge`, `FAQ_Yield_Unrealistic`, `FAQ_Report_Variables_Empty`
4. Email instructor with:
    - .apsimx file (or key excerpts)
    - Error message (screenshot)
    - What you were trying to do (steps)
    - Expected vs. observed behavior

# References

# Bibliography


[1] APSIM Initiative (2025). *APSIM Next Gen Documentation*. https://www.apsim.info

[2] Holzworth, D.P., Huth, N.I., deVoil, P.G., et al. (2014). APSIM — Evolution towards a new generation of agricultural systems simulation. *Environmental Modelling & Software*, 62, 327–350. https://doi.org/10.1016/j.envsoft.2014.07.009

[3] Keating, B.A., Carberry, P.S., Hammer, G.L., et al. (1998). An overview of APSIM, a model designed for farming systems simulation. *European Journal of Agronomy*, 18, 267–288.

# LLM Wiki Resources

Throughout this tutorial, we reference concepts stored in the SCIE3314 APSIM Wiki:

**Entities** Definition pages for APSIM components (Clock, Weather, Soil, Manager, Report, DataStore, Wheat, etc.) with misconceptions and troubleshooting.

**Concepts** Pedagogic synthesis pages (Scaffolding, Near-Solved Design, Water Stress, Phenology-Thermal-Time Link, NUE, Module Dependencies).

**FAQ** Diagnostic pages for common errors and decisions.

**Exercises** Step-by-step guides for Exercises A–D with learning objectives and analysis prompts.

Wiki location: `https://github.com/00113844/SCIE3314_APSIM_Tutorial/tree/main/SCIE3314_APSIM_Wiki`

# Further Resources

- **APSIM Next Gen documentation and tutorials:**
  https://apsimnextgeneration.netlify.app/user_tutorials/
- **APSIM community forum and support:**
  https://www.apsim.info/
- **ApsimX source code (GitHub):**
  https://github.com/APSIMInitiative/ApsimX
- Holzworth, D.P. et al. (2014). APSIM — Evolution towards a new generation of agricultural systems simulation. *Environmental Modelling & Software*, 62, 327–350. https://doi.org/10.1016/j.envsoft.2014.07.009

# Instructor Notes

- Partial `.apsimx` exercise files (A–D) should be distributed via LMS *before* the first session.
- All four exercises require APSIM Next Gen $\geq$ version 2023.2.7164.0.
- Expected run times: Exercises A, B, C <30 seconds each; Exercise D ≈1–2 minutes.
- Exercise C requires the `Experiment` node — ensure students have completed Module 1 (component reference) before starting.
- Exercises B and D are suitable for individual assessment: the data tables in each exercise can be used as submission items.
- Required files: `wheat_fallow_partial.apsimx`, `wheat_Nfert_partial.apsimx`, `wheat_sowingdate_partial.apsimx`, `wheat_whatif_partial.apsimx` (provided via LMS).